
Control System Fault Codes

Control System

How to Retrieve Control System Fault Codes

The machine control system contains two different types of fault codes

Active Faults: These faults have been detected in the current operating session. The presence of an active fault will be reflected by the System Fault Indicator lights and the alarm.

Logged Faults: The faults contained in the fault log have occurred in the past and may or may not still be in effect.

Both faults may be viewed by pressing the Fault Button above the ground display and navigating to either the Active Machine Faults screen or the Machine Fault Log screen. The list of faults will be shown by fault code and the hour meter of when it occurred.

The fault lists can be browsed. To learn more about a fault entry, select it and press OK. A screen will be shown that describes the fault in more detail.

Control System Fault Codes

Models with Revision G Software or Lower

ID	Component	ID	Name	Cause	Effect	Solution
6	Keyswitch	19	Fault	Signals C134PLT and C174GND from KS1 are both HIGH for at least 1 second, causing both control stations to be selected.	Machine will not be functional.	Verify with a voltmeter that signals switch properly. * With the keyswitch in the Ground position, C174GND (WH, TCON J5-12) should be 12 V and C134PLT (GR/BK, TCON J5-8) should be 0 V. * With the keyswitch in the Platform position, C174GND (WH, TCON J5-12) should be 0 V and C134PLT (GR/BK, TCON J5-8) should be 12 V. Cycle power to allow the machine to re-test the fault.
11	Engine Start Toggle Switch	19	Stuck ON at power up.	Toggle switch input was detected as HIGH at start-up.	Engine start switch will not operate.	Verify with a voltmeter that the Engine Start Toggle Switch (TS2) is operating properly. * C33STR (BK, PCON J16-3) should be 0 V with the switch OFF and 12 V with the switch ON. Cycle power to allow the machine to re-test the fault.
12	CAN Bus	29	System not functioning	CAN bus system is not functioning properly.	Machine will not be functional.	Verify CAN bus is functional. Pull out connector J5 from the TCON and with a voltmeter set to resistance verify that pins 3 and 4 on the harness connector measure between 55 and 65 Ohms and that neither pin has a short to power or ground. Check wiring and boom cable for open or short circuits if resistance check fails.
13	Jib Up/Down Toggle Switch	19	Stuck ON at power up.	Toggle switch input was detected as HIGH at start-up.	The Jib Up or Down functions will not operate.	Verify with a voltmeter that the Jib Up/Down Toggle Switch (TS8) is operating properly. * C43JU (BL, PCON J17-6) should be 0 V with the switch OFF and 12 V with the switch held in the UP position. * C44JD (BL/BK, PCON J17-8) should be 0 V with the switch OFF and 12 V with the switch held in the DOWN position. Cycle power to allow the machine to re-test the fault.
14	Battery Voltage	15	Voltage is low.	Battery voltage is too low while engine is running.	Advisory fault. Machine is fully functional.	Verify with a voltmeter that the alternator is operating properly. * With the engine running, battery voltage should be above 12.3 V. * Verify alternator and power wiring. * Replace alternator if necessary. * Replace battery if necessary.

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15	Foot Switch	19	Stuck ON at power up, or timeout occurred.	Foot switch input was detected as HIGH at start-up, or the footswitch was held for too long and caused the foot switch timer to elapse.	Machine will not be functional.	* Release foot switch and cycle power. * If problem persists, with a voltmeter verify that C24FS (WH, PCON J12-2) is LOW when foot switch is released and 12 V when held down.
19	APU Solenoid and Switch	12	Solenoid output shorted to V+	TCON detected that C27AUX circuit is shorted to 12 V.	Machine will not be functional.	Check C27AUX (RD, TCON J3-3) wiring. * With a voltmeter verify that C27AUX (RD, TCON J3-3) is 0 V when the Auxiliary Power toggle switch is OFF and 12 V when switch is ON. * Verify the terminals of the APU solenoid have a resistance greater than 5 ohms. If not, replace the solenoid.
		15	Solenoid output shorted to GND	TCON detected that C27AUX circuit is shorted to GND.	Auxiliary Power function will not operate.	
		19	Stuck ON at power up.	Toggle switch input was detected as HIGH at start-up.	Auxiliary Power function will not operate from platform.	Verify with a voltmeter that the AUX Power Switch (TS1) is operating properly. * C47AUX (RD, PCON J17-10) should be 0 V with the switch OFF and 12 V with the switch ON. Cycle power to allow the machine to re-test the fault.
20	Primary Boom Extend / Retract Joystick	12	Joystick voltage too high.	Circuit C163PER is above 4.5 V.	Boom Extend and Retract functions will not operate.	* Leave joystick in neutral position and cycle power. * If fault persists, with a voltmeter check that C163PER (BL/WH, PCON J18-4) is at 2.5V +/- 0.2 V at startup and does not go above 4.5 V or below 0.5 V when fully stroked in each direction. * Replace joystick if voltage check fails.
		15	Joystick voltage too low.	Circuit C163PER is below 0.5 V.		
		19	Joystick held at startup.	Circuit C163PER voltage was not within the startup voltage range of 2.5 V +/- 0.2 V.		

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21	Primary Boom Up / Down Joystick	12	Joystick voltage too high.	Circuit C164PUD is above 4.5 V.	Boom Up and Down functions will not operate.	* Leave joystick in neutral position and cycle power. * If fault persists, with a voltmeter check that C164PUD (RD/WH, PCON J18-3) is at 2.5V +/- 0.2 V at startup and does not go above 4.5 V or below 0.5 V when fully stroked in each direction. * Replace joystick if voltage check fails.
		15	Joystick voltage too low.	Circuit C164PUD is below 0.5 V.		
		19	Joystick held at startup.	Circuit C164PUD voltage was not within the startup voltage range of 2.5 V +/- 0.2 V.		
22	Primary Boom Up Valve	12	Circuit shorted to V+	TCON detected that C1PBU circuit is shorted to 12 V.	Boom Up function will not operate. Boom Down will also not operate if short is to 12 V.	* With a voltmeter, measure Y22. Proper resistance should be between 5 - 60 ohms. Replace Y22 if resistance is outside of those values. * Verify wiring for C1PBU (RD, TCON J4-10) and check for open or short circuits.
		15	Circuit shorted to GND	TCON detected that C1PBU circuit is shorted to GND.		
23	Primary Boom Down Valve	12	Circuit shorted to V+	TCON detected that C2PBD circuit is shorted to 12 V.	Boom Up and Down functions will not operate.	* With a voltmeter, measure Y21. Proper resistance should be between 5 - 60 ohms. Replace Y21 if resistance is outside of those values. * Verify wiring for C2PBD (RD/BK, TCON J4-9) and check for open or short circuits. * Verify resistance between Recovery Module pins J37-2 and J37-11 is zero ohms.
		15	Circuit shorted to GND	TCON detected that C2PBD circuit is shorted to GND.		

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24	Boom Angle Sensor	13	No signal	No CAN messages have been received by the TCON from the Angle Sensor.	Boom Up, Extend and Turntable Rotate functions will not operate.	* Verify sensor power P85PWR (GR, TCON J6-7) is at 12 V while the system is running. * Verify the sensor is receiving power. Unplug the sensor and verify 12 V across pins 2 (+) and 1 (-) on the harness connector. * Verify the CAN bus wiring. With the sensor still unplugged, measure resistance across harness connector pins 3 and 4. If resistance is not between 55 and 65 ohms, check CAN wire harnesses.* Replace sensor if necessary.
		17	Invalid calibration	Sensor has been replaced and requires calibration, or calibration in memory has been erased.		Run the boom angle calibration process and verify fault is no longer active.
		19	Out of range	Sensor disagrees with boom position sensed by LST3S, or boom exceeded the calibrated minimum or maximum values.		* Run the boom angle calibration process and verify fault is no longer active. * If fault persists, verify LST3S (C68LS, RD, TCON J7-3) operates properly. The signal should be 12 V when the boom is stowed and 0 V when the boom is up.
		36	Internal check	The two internal sensing channels inside the Angle Sensor have registered a significant difference.		* Run the boom angle calibration process and verify fault is no longer active. * If fault persists, replace the sensor.
25	Cable tension	21	Loose cable	Boom cable tension limit switch LSB2S indicates that the tension is loose on the boom cables.	Boom Up and Extend functions will not operate.	* Return boom to stowed position as quickly as possible. * Verify both boom cables are properly tensioned and that LST3S is properly centered in the bracket. * Verify LSB2S (C59CNK, BL/WH, TCON J7-5) operates properly. Signal should normally be 12 V. If cable tension is loose, signal should be 0 V.

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26	Primary Boom Extend Valve	12	Circuit shorted to V+	TCON detected that C7PBE circuit is shorted to 12 V.	Boom Extend and Retract functions will not operate.	* With a voltmeter, measure Y26. Proper resistance should be between 5 - 60 ohms. Replace Y26 if resistance is outside of those values. * Verify wiring for C7PBE (BK, TCON J4-7) and check for open or short circuits.
		15	Circuit shorted to GND	TCON detected that C7PBE circuit is shorted to GND.		
27	Primary Boom Retract Valve	12	Circuit shorted to V+	TCON detected that C8PBR circuit is shorted to 12 V.	Boom Extend and Retract functions will not operate.	* With a voltmeter, measure Y25. Proper resistance should be between 5 - 60 ohms. Replace Y25 if resistance is outside of those values. * Verify wiring for C8PBR (BK/WH, TCON J4-8) and check for open or short circuits. * Verify resistance between Recovery Module pins J37-5 and J37-8 is zero ohms.
		15	Circuit shorted to GND	TCON detected that C8PBR circuit is shorted to GND.		
35	Ground Buttons	19	Held at startup	Buttons were held at system start, or are shorted on the ground control panel.	The function associated with the stuck button will not operate.	* Release the ground control buttons and cycle power. * If problem persists, open ground control box and open the Display Controller access door, being careful to keep water out. Verify the ribbon cables are properly seated and dry. * Re-seat and/or replace ribbon cables if necessary. * Replace ground control overlay.
36	Jib Up Valve	12	Circuit shorted to V+	PCON detected that V43JU circuit is shorted to 12 V.	Jip Up function will not operate.	* With a voltmeter, measure Y68. Proper resistance should be between 5 - 60 ohms. Replace Y68 if resistance is outside of those values. * Verify wiring for V43JU (BL, PCON J12-8) and check for open or short circuits.
		15	Circuit shorted to GND	PCON detected that V43JU circuit is shorted to GND.		
37	Jib Down Valve	12	Circuit shorted to V+	PCON detected that V44JD circuit is shorted to 12 V.	Jib Down function will not operate.	* With a voltmeter, measure Y69. Proper resistance should be between 5 - 60 ohms. Replace Y69 if resistance is outside of those values. * Verify wiring for V44JD (BL/BK, PCON J12-9) and check for open or short circuits.
		15	Circuit shorted to GND	PCON detected that V44JD circuit is shorted to GND.		

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41	Turntable Rotate Joystick	12	Joystick voltage too high.	Circuit C165TTR is above 4.5 V.	Turntable Rotate functions will not operate.	* Leave joystick in neutral position and cycle power. * If fault persists, with a voltmeter check that C165TTR (WH/RD, PCON J18-2) is at 2.5V +/- 0.2 V at startup and does not go above 4.5 V or below 0.5 V when fully stroked in each direction. * Replace joystick if voltage check fails.
		15	Joystick voltage too low.	Circuit C165TTR is below 0.5 V.		
		19	Joystick held at startup.	Circuit C165TTR voltage was not within the startup voltage range of 2.5 V +/- 0.2 V.		
42	Turntable Rotate CW Valve	12	Circuit shorted to V+	TCON detected that C4TRL circuit is shorted to 12 V.	Turntable Rotate functions will not operate.	* With a voltmeter, measure Y24. Proper resistance should be between 5 - 60 ohms. Replace Y24 if resistance is outside of those values. * Verify wiring for C4TRL (WH, TCON J3-12) and check for open or short circuits.
		15	Circuit shorted to GND	TCON detected that C4TRL circuit is shorted to GND.		
43	Turntable Rotate CCW Valve	12	Circuit shorted to V+	TCON detected that C5TRR circuit is shorted to 12 V.	Turntable Rotate functions will not operate.	* With a voltmeter, measure Y23. Proper resistance should be between 5 - 60 ohms. Replace Y23 if resistance is outside of those values. * Verify wiring for C5TRR (WH/BK, TCON J3-1) and check for open or short circuits.
		15	Circuit shorted to GND	TCON detected that C5TRR circuit is shorted to GND.		
44	Drive Enable Switch	19	Stuck ON at power up.	Toggle switch input was detected as HIGH at start-up.	Drive Enable switch will not operate, possibly preventing drive while the turntable is rotated.	Verify with a voltmeter that the Drive Enable Toggle Switch (TS15) is operating properly. * C143DEL (BL/RD, PCON J16-7) should be 0 V with the switch OFF and 12 V with the switch held in the ON position. Cycle power to allow the machine to re-test the fault.
45	Platform Level Up/Down Switch	19	Stuck ON at power up.	Toggle switch input was detected as HIGH at start-up.	Platform Level Up or Down functions may not operate, depending on which direction the switch fault occurred.	Verify with a voltmeter that the Platform Level Up/Down Toggle Switch (TS9) is operating properly. * C14PLU (OR, PCON J17-2) should be 0 V with the switch OFF and 12 V with the switch held in the UP position. * C15PLD (OR/BK, PCON J17-3) should be 0 V with the switch OFF and 12 V with the switch held in the DOWN position. Cycle power to allow the machine to re-test the fault.

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ID	Component	ID	Name	Cause	Effect	Solution
46	Platform Rotate CW Valve	12	Circuit shorted to V+	PCON detected that V17PRL circuit is shorted to 12 V.	Platform Rotate CW function will not operate.	* With a voltmeter, measure Y70. Proper resistance should be between 5 - 60 ohms. Replace Y70 if resistance is outside of those values. * Verify wiring for V17PRL (GR, PCON J12-5) and check for open or short circuits.
		15	Circuit shorted to GND	PCON detected that V17PRL circuit is shorted to GND.		
47	Platform Rotate CCW Valve	12	Circuit shorted to V+	PCON detected that V18PRR circuit is shorted to 12 V.	Platform Rotate CCW function will not operate.	* With a voltmeter, measure Y71. Proper resistance should be between 5 - 60 ohms. Replace Y71 if resistance is outside of those values. * Verify wiring for V18PRR (GR/BK, PCON J12-4) and check for open or short circuits.
		15	Circuit shorted to GND	PCON detected that V18PRR circuit is shorted to GND.		
51	Drive Forward / Reverse Joystick	12	Joystick voltage too high.	Circuit C160DRV is above 4.5 V.	Drive Forward and Reverse functions will not operate.	* Leave joystick in neutral position and cycle power. * If fault persists, with a voltmeter check that C160DRV (WH/RD, PCON J18-10) is at 2.5 V +/- 0.2 V at startup and does not go above 4.5 V or below 0.5 V when fully stroked in each direction. * Replace joystick if voltage check fails.
		15	Joystick voltage too low.	Circuit C160DRV is below 0.5 V.		
		19	Joystick held at startup.	Circuit C160DRV voltage was not within the startup voltage range of 2.5 V +/- 0.2 V.		
52	Drive Forward Valve	12	Circuit shorted to V+	TCON detected that C30FWD circuit is shorted to 12 V.	Drive Forward, Brake Release and possibly Engine will not operate.	* With a voltmeter, measure Y6. Proper resistance should be between 5 - 60 ohms. Replace Y6 if resistance is outside of those values. * Verify wiring for C30FWD (WH, TCON J2-6) and check for open or short circuits.
		15	Circuit shorted to GND	TCON detected that C30FWD circuit is shorted to GND.		
53	Drive Reverse Valve	12	Circuit shorted to V+	TCON detected that C31REV circuit is shorted to 12 V.	Drive Reverse, Brake Release and possibly Engine will not operate.	* With a voltmeter, measure Y5. Proper resistance should be between 5 - 60 ohms. Replace Y5 if resistance is outside of those values. * Verify wiring for C31REV (WH/BK, TCON J2-7) and check for open or short circuits.
		15	Circuit shorted to GND	TCON detected that C31REV circuit is shorted to GND.		

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ID	Component	ID	Name	Cause	Effect	Solution
54	Brake Release Valve	12	Circuit shorted to V+	TCON detected that C32BRK circuit is shorted to 12 V.	Machine will not be functional.	* With a voltmeter, measure Y2. Proper resistance should be between 5 - 60 ohms. Replace Y2 if resistance is outside of those values. * Verify wiring for C32BRK (WH/RD, TCON J2-9) and check for open or short circuits.
		15	Circuit shorted to GND	TCON detected that C32BRK circuit is shorted to GND.		
55	Motor Stroke Valve	12	Circuit shorted to V+	TCON detected that C29MS circuit is shorted to 12 V.	Drive functions will not operate.	* With a voltmeter, measure Y27. Proper resistance should be between 5 - 60 ohms. Replace Y27 if resistance is outside of those values. * Verify wiring for C29MS (RD/WH, TCON J2-8) and check for open or short circuits.
		15	Circuit shorted to GND	TCON detected that C29MS circuit is shorted to GND.		
61	Steer Joystick	12	Joystick voltage too high.	Circuit C159STD or C159STA is above 4.5 V.	Drive and steer functions will not operate.	* Leave joystick in neutral position and cycle power. * If fault persists, identify which circuit is the steer circuit. Check C159STA (BL/WH, PCON J18-11) for a joystick steer option. Check C159STD (BL/RD, PCON J18-9) for the rocker joystick steer option. With a voltmeter check that the circuit is at 2.5V +/- 0.2 V at startup and does not go above 4.5 V or below 0.5 V when fully stroked in each direction. * Replace joystick if voltage check fails.
		15	Joystick voltage too low.	Circuit C159STD or C159STA is below 0.5 V.		
		19	Joystick held at startup.	Circuit C159STD or C159STA voltage was not within the startup voltage range of 2.5 V +/- 0.2 V.		
68	Safety Envelope Fault	21	Out of range	The system has determined that the platform position has exceeded the safety envelope boundary.	All functions will not operate except for the recovery procedure.	Use Auxiliary Power and the boom retract and/or down functions to bring the platform back within the operational envelope.
69	Generator Control Circuit	12	Circuit shorted to V+	TCON detected that C45GEN circuit is shorted to 12 V.	Generator function will not operate.	* With a voltmeter, measure Y29. Proper resistance should be between 5 - 60 ohms. Replace Y29 if resistance is outside of those values. * Verify wiring for C45GEN (GR/WH, TCON J3-2) and check for open or short circuits.
		15	Circuit shorted to GND	TCON detected that C45GEN circuit is shorted to GND.		

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ID	Component	ID	Name	Cause	Effect	Solution
71	ESTOP Circuit	15	ESTOP Return circuit shorted to GND	Mismatched signals detected on ESTOP Return circuits. Either: * PCON ESTOP 2 and ESTOP 4 do not match, or * TCON ESTOP 2 and ESTOP 4 do not match.	ESTOP circuit cannot complete, preventing machine functions.	* Inspect PCON J15 pins 11 and 12 for open or short circuits. Using a voltmeter, set to diode test mode and make sure the diode tests ok with the anode at pin 12 and cathode at pin 11. If test fails, replace the PCON. * Inspect TCON J5 pins 11 and 12 for open or short circuits. Using a voltmeter, set to diode test mode and make sure the diode tests ok with the anode at pin 12 and cathode at pin 11. If test fails, replace the PCON. * Check P23RET (BK) for shorts to ground or power.
		16	ESTOP Power circuit shorted to GND	TCON detected that P56PWR is shorted to GND, thereby preventing the ESTOP circuit from functioning.		Check P56PWR (RD, TCON J5-7) for open or short circuits. Verify power is reaching the P1 ESTOP contact block. When ESTOP is pulled out, verify power is reaching the KS1 keyswitch contact blocks.
		21	ESTOP Return circuit mismatch	TCON detected that ESTOP Return circuits are not matching, indicating a fault with either P23PWR or P23RET.		* Check for open and short circuits in the boom cable. * Verify that there is low resistance (less than five ohms) on the cabling between TCON J5-11 and PCON J15-11 (BK wire in harness and boom cable). An open circuit between these pins indicates a harness or cable fault. * Verify that there is low resistance (less than five ohms) on the cabling between TCON J5-10 and PCON J15-10 (RD wire in harness and boom cable). An open circuit between these pins indicates a harness or cable fault.
		22	ESTOP Source circuit is shorted to GND	TCON detected that ESTOP 1 is at 12 V but ESTOP 3 is at 0 V.		* Using a voltmeter and with the system turned OFF, check the resistance between TCON J5 pins 9 and 10. Replace TCON if resistance is more than 5 ohms.
		31	ESTOP Source circuit mismatch	TCON detected that ESTOP 1 is at 0 V but ESTOP 3 is at 12 V.		* Check P79PWR (BK, TCON J5-9) and P23PWR (RD, TCON J5-10) for open or short circuits.

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72	Contact Alarm	36	Switch state mismatch	PCON detected that inputs from SW30 (contact alarm proximity sensor) are the same voltage.	Boom Up and Extend functions will not operate.	* Using a voltmeter, verify that the voltage on C248CA (OR/BK, PCON J12-1) is always the opposite of C247CA (OR, PCON J12-11). * Check wiring for open or short circuits. * Verify SW30 is receiving power and ground by verifying 12 V across C31 pins 1 and 3. * Replace SW30 if necessary.
75	TCON Secondary Processor Software	34	Incorrect software version	The TCON's secondary processor software version does not match the system software revision level.	Boom Up and Extend functions will not operate.	Reprogram the machine with latest software. Check http://firmware.genielift.com/ for the latest software version.
76	Platform Load Sensor	13	No signal	No CAN messages have been received by the TCON from the platform load sensor.	Boom Up, Extend and Turntable Rotate functions will not operate.	* Verify sensor power P26PWR (BK, PCON J13-9) is at 12 V while the system is running. * Verify 12 V across pins on C27-2 (+) and C27-3 (-) on the harness connector. * Measure resistance across C27 pins 4 and 5 on the harness connector. If resistance is not between 55 and 65 ohms, check CAN wire harnesses. * Replace sensor if necessary.
		15	Reverse loading	The sensor has reported a reversed or negative load condition. This fault is triggered when the boom up function has operated for at least 10 seconds and the platform load is consistently detected as less than -220 lbs.		* Make sure the platform is not touching the ground or an overhead obstruction. Cycle power and check that fault is no longer reported. * Perform the platform overload calibration procedure. * Replace sensor if necessary.

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ID	Component	ID	Name	Cause	Effect	Solution
76		17	Invalid calibration	The platform load sensor has no valid calibration data, either because the sensor was replaced or the prior calibration was cleared from memory or was unsuccessful.		* Run the platform overload calibration process and verify fault is no longer active.
		21	Internal check	The two internal sensing channels inside the Angle Sensor have registered a significant difference.		* Run the platform overload calibration process and verify fault is no longer active. * If fault persists, replace the sensor.

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77	Platform Overload Detection	21	Internal check	The secondary platform overload check detected overload before the operational platform overload check.	All functions will not operate except for the recovery procedure.	* Cycle power and verify condition clears itself. * If problem persists, run the platform overload calibration procedure. * If problem persists, replace the sensor.
78	Platform Controller (PCON)	13	Not detected	TCON has not detected the PCON primary processor on the communications bus.	Controls from the platform will not operate.	* Verify PCON is receiving power. Check for 12 V on J15-2 (+) and J15-1 (-). * Verify PCON is receiving ESTOP signals. With ground ESTOP pulled out and keyswitch set to Platform, verify 12 V is reaching PCON J15-10. * Verify CAN bus is functional. Pull out connector J15 from the PCON and with a voltmeter set to resistance verify that pins 3 and 4 on the harness connector measure between 110 and 130 Ohms. Check wiring and boom cable for open or short circuits if resistance check fails. * Replace PCON if necessary.
		34	Incorrect software version	PCON software revision does not match the system revision level.	Boom Up and Extend functions will not operate.	Reprogram the machine with latest software. Check http://firmware.genielift.com/ for the latest software version.
		36	Secondary processor not detected.	TCON has not detected the PCON secondary processor on the communications bus.	Controls from the platform will not operate.	Reprogram the machine with latest software. Check http://firmware.genielift.com/ for the latest software version.
79	Display Controller (DISCON)	13	Not detected	TCON has not detected the DISCON on the communications bus.	Controls from the ground will not operate.	* Verify DISCON is operating by checking that the display on the front is operating properly. * If there is no display, try to power up the DISCON by setting the keyswitch to Ground, pull out the ground ESTOP and press the Ground Function Enable button. The display should start. * If there is still no display, check DISCON power on J10 pins 2 (+) and 1 (-). * If DISCON operates but fault persists, check the CAN bus wiring. Disconnect J10 and with a voltmeter set to resistance verify that the resistance between pins 3 and 4 on the harness connector are between 55 and 65 ohms. Check CAN wiring if resistance check fails. * Replace DISCON if required.
		34	Incorrect software version	DISCON software revision does not match the system revision level.	Boom Up and Extend functions will not operate.	Reprogram the machine with latest software. Check http://firmware.genielift.com/ for the latest software version.

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80	Turntable Controller (TCON)	13	Not detected	The DISCON has not detected the TCON on the CAN bus.	Machine will not be functional.	* Verify TCON is receiving power. Check for 12 V on J5-2 (+) and J5-1 (-). * Verify TCON is receiving ESTOP signals. With ground ESTOP pulled out and keyswitch set to Ground, verify 12 V is reaching TCON J5-9 and J5-12. * Verify CAN bus is functional. Pull out connector J5 from the TCON and with a voltmeter set to resistance verify that pins 3 and 4 on the harness connector measure between 55 and 65 Ohms. Check wiring and boom cable for open or short circuits if resistance check fails. * Replace TCON if necessary.
		36	Incorrect software version	TCON's secondary processor software revision does not match the system revision level.		Reprogram the machine with latest software. Check http://firmware.genielift.com/ for the latest software version.
83	Telematics	21	Remote disable	The telematics module has locked out machine operation.	Machine will not be functional, or have limited functionality, depending on the telematics option.	Call machine owner to re-enable machine functions.
84	Turntable Tilt Sensor	13	Not detected	The tilt sensors (S8O and/or S8S) have not been detected on the CAN bus.	Boom Up, Extend and Drive functions will not operate.	* Verify S8O is receiving power. Disconnect sensor connector and verify harness connector pins 1 and 2 measure 12 V. * Verify S8S is receiving power. Disconnect sensor connector and verify harness connector pins 1 and 2 measure 12 V. * Verify CAN bus wiring. With a voltmeter set to resistance, check the resistance between pins 3 and 4 on the harness connector (not the sensor's integrated connector). Resistance should be between 55 and 65 ohms. Check CAN wiring if resistance check fails. * Replace sensor if required.
		17	Invalid calibration	The tilt sensors have no valid calibration data, either because the sensor(s) were replaced or the prior calibration was cleared from memory or was unsuccessful.	Boom Up and Extend functions will not operate.	Perform the tilt sensor calibration process.

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88	Function Enable Valve	12	Circuit shorted to V+	TCON detected that C155PCE circuit is shorted to 12 V.	Only Auxiliary Power is available.	* With a voltmeter, measure Y74. Proper resistance should be between 5 - 60 ohms. Replace Y74 if resistance is outside of those values. * Verify wiring for C155PCE (OR/RD, TCON J2-5) and check for open or short circuits.
		15	Circuit shorted to GND	TCON detected that C155PCE circuit is shorted to GND.		
92	Turntable Rotate Flow Valve	12	Circuit shorted to V+	TCON detected that C6TRF circuit is shorted to 12 V.	Turntable Rotate functions will not operate.	* With a voltmeter, measure Y13. Proper resistance should be between 5 - 60 ohms. Replace Y13 if resistance is outside of those values. * Verify wiring for C6TRF (WH/RD, TCON J4-1) and check for open or short circuits.
		15	Circuit shorted to GND	TCON detected that C6TRF circuit is shorted to GND.		
94	Boom Length Sensor	13	No signal	No CAN messages have been received by the TCON from the Length Sensor.	Boom Up, Extend and Turntable Rotate functions will not operate.	* Verify sensor power P85PWR (GR, TCON J6-7) is at 12 V while the system is running. * Verify the sensor is receiving power. Unplug the sensor and verify 12 V across pins 1 (+) and 2 (-) on the harness connector. * Verify the CAN bus wiring. With the sensor still unplugged, measure resistance across harness connector pins 3 and 4. If resistance is not between 55 and 65 ohms, check CAN wire harnesses. * Replace sensor if necessary.
		17	Invalid calibration	Sensor has been replaced and requires calibration, or calibration in memory has been erased.		Run the boom length calibration process and verify fault is no longer active.
		19	Out of range	Sensor disagrees with boom position sensed by LSB1RS, or boom exceeded the calibrated minimum or maximum values.		* Run the boom length calibration process and verify fault is no longer active. * If fault persists, verify LSB1RS C42LS, OR/RD, Recovery Module J36-4 and J36-9, TCON J7-4) operates properly. The signal should be 12 V when the boom is fully retracted and 0 V when the boom is extended at least 24 inches.
		36	Internal check	The two internal sensing channels inside the Length Sensor have registered a significant difference.		* Run the boom length calibration process and verify fault is no longer active. * If fault persists, replace the sensor.

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
96	Engine Oil Pressure	21	Value too low	The engine oil pressure is measured as too low, potentially damaging the engine.	Engine may not attempt to start until system power cycle. Engine may be turned off by control system immediately.	* Stop engine as soon as possible. * Verify the engine oil level. * Verify the engine oil pump. * Power on the machine and start the engine. At the ground controls display, view the engine oil pressure gauge for proper operation. * If engine uses an analog oil pressure sensor, verify it is sending an analog voltage on C25PSR (WH/BK, TCON J6-9).
100	Platform Level Up Valve	12	Circuit shorted to V+	PCON detected that V14PLU circuit is shorted to 12 V.	Platform Level Up and possibly Level Down functions will not operate.	* With a voltmeter, measure Y20. Proper resistance should be between 5 - 60 ohms. Replace Y20 if resistance is outside of those values. * Verify wiring for V14PLU (OR, PCON J12-6) and check for open or short circuits.
		15	Circuit shorted to GND	PCON detected that V14PLU circuit is shorted to GND.		
101	Platform Level Down Valve	12	Circuit shorted to V+	PCON detected that V15PLD circuit is shorted to 12 V.	Platform Level Down and possibly Level Up functions will not operate.	* With a voltmeter, measure Y19. Proper resistance should be between 5 - 60 ohms. Replace Y19 if resistance is outside of those values. * Verify wiring for V15PLD (OR/BK, PCON J12-7) and check for open or short circuits.
		15	Circuit shorted to GND	PCON detected that V15PLD circuit is shorted to GND.		
150	Platform Overload Recovery	21	Condition occurred	This is a flag that indicates that the platform was overloaded and that auxiliary function enable was used while overloaded.	Advisory condition. Machine is fully functional.	Power up the machine in service mode and clear the flag in the service menu on the ground control display.
152	Primary Boom Up/Down Flow Valve	12	Circuit shorted to V+	TCON detected that C3PBF circuit is shorted to 12 V.	Boom Up and Down functions will not operate.	* With a voltmeter, measure Y12. Proper resistance should be between 5 - 60 ohms. Replace Y12 if resistance is outside of those values. * Verify wiring for C3PBF (RD/WH, TCON J4-4) and check for open or short circuits. * Verify resistance between Recovery Module pins J37-3 and J37-10 is zero ohms.
		15	Circuit shorted to GND	TCON detected that C3PBF circuit is shorted to GND.		

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
160	Steer Left Valve	12	Circuit shorted to V+	TCON detected that C37STL circuit is shorted to 12 V.	Steer Left function will not operate.	* With a voltmeter, measure Y3. Proper resistance should be between 5 - 60 ohms. Replace Y3 if resistance is outside of those values. * Verify wiring for C37STL (BL/BK, TCON J4-11) and check for open or short circuits.
		15	Circuit shorted to GND	TCON detected that C37STL circuit is shorted to GND.		
161	Steer Right Valve	12	Circuit shorted to V+	TCON detected that C36STR circuit is shorted to 12 V.	Steer Right function will not operate.	* With a voltmeter, measure Y4. Proper resistance should be between 5 - 60 ohms. Replace Y4 if resistance is outside of those values. * Verify wiring for C36STR (BL, TCON J4-12) and check for open or short circuits.
		15	Circuit shorted to GND	TCON detected that C36STR circuit is shorted to GND.		
169	Primary Boom Extend / Retract Flow Valve	12	Circuit shorted to V+	TCON detected that C9PEF circuit is shorted to 12 V.	Boom Extend and Retract functions will not operate.	* With a voltmeter, measure Y15. Proper resistance should be between 5 - 60 ohms. Replace Y15 if resistance is outside of those values. * Verify wiring for C9PEF (BK/RD, TCON J4-5) and check for open or short circuits. * Verify resistance between Recovery Module pins J37-6 and J37-7 is zero ohms.
		15	Circuit shorted to GND	TCON detected that C9PEF circuit is shorted to GND.		
174	Platform Rotate Toggle Switch	19	Stuck ON at power up.	Toggle switch input was detected as HIGH at start-up.	Platform Rotate Left or Right will not operate, depending on which side of the switch was faulted.	Verify with a voltmeter that the Platform Rotate Toggle Switch (TS7) is operating properly. * C17PRL (GR, PCON J16-4) and C18PRR (GR/BK, PCON J17-5) should be 12 V when the switch is pressed in their respective directions. Cycle power to allow the machine to re-test the fault.
177	Engine Coolant Temperature	12	Value too high	The engine coolant temperature is measured as too high, potentially damaging the engine.	Engine will not restart until machine power is cycled. Engine may stop itself, depending on the engine option.	* Stop engine as soon as possible. * Verify the engine coolant level. * Verify the engine coolant/water pump. * Power on the machine and start the engine. At the ground controls display, view the engine temperature gauge for proper operation. * If engine uses an analog temperature sensor, verify it is sending an analog voltage on C26TSR (WH/RD, TCON J6-10).

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
178	Engine Warning Lamp	21	Fault	The engine has triggered the Engine Warning Lamp condition.	Machine will function as normal.	* Check the engine fault codes with an engine diagnostic tool.
179	Engine Stop Lamp	21	Fault	The engine has triggered the Engine Stop Lamp condition.	Control system will not allow engine restart until power cycle. Engine may stop itself, depending on the engine option.	* Stop engine immediately. * Check the engine fault codes with an engine diagnostic tool.
240	Sensor Power P85PWR	12	Circuit shorted to V+	TCON detected that P85PWR circuit is shorted to 12 V.	Boom Up and Extend functions will not operate.	* Turn on machine and verify sensor power P85PWR (GR, TCON J6-7) is at 12 V. * Inspect wiring for open or shorts circuits.
		15	Circuit shorted to GND	TCON detected that P85PWR circuit is shorted to GND.		
241	Sensor Power P109PWR	12	Circuit shorted to V+	TCON detected that P109PWR circuit is shorted to 12 V.	Boom Up and Extend functions will not operate.	* Turn on machine and verify sensor power P109PWR (GR/WH, TCON J7-7) is at 12 V. * Inspect wiring for open or shorts circuits.
		15	Circuit shorted to GND	TCON detected that P109PWR circuit is shorted to GND.		
242	Sensor Power P26PWR	12	Circuit shorted to V+	PCON detected that P26PWR circuit is shorted to 12 V.	Boom Up and Extend functions will not operate.	* Turn on machine and verify sensor power P26PWR (BK, PCON J13-9) is at 12 V. * Inspect wiring for open or shorts circuits.
		15	Circuit shorted to GND	PCON detected that P26PWR circuit is shorted to GND.		
243	Proximity Override Toggle Switch	19	Stuck ON at power up.	Toggle switch input was detected as HIGH at start-up.	Operator will be unable to override the proximity cutoff feature.	Verify with a voltmeter that the Proximity Override Toggle Switch (TS46) is operating properly. * C90PRX (RD/BK, PCON J17-9) should be 12 V when the switch is pressed. Cycle power to allow the machine to re-test the fault.

Control System Fault Codes

Models with Revision H software or higher

ID	Component	ID	Name	Cause	Effect	Solution
1	Engine Can Bus	13	Not detected	Engine CANBUS CAN+ or CAN- shorted to supply.	Disable engine start for tier 4F engines	*Verify with a voltmeter that CAN+ and CAN- lines are not shorted.* Measure the resistance between D228CAN+ (YL, TCON J9-3) and D228CAN- (GR, TCON J9-4); should be between 55 and 65 ohms. * D228CAN+ (YL, TCON J9-3) should be between 1V and 5V. * D228CAN- (GR, TCON J9-4) should be between 1V and 5V. Remove short(s) and cycle power.
7	Engine Warning	21	Fault	The engine has triggered the Engine Warning Lamp condition.	Machine will function as normal.	* Check the engine fault codes with an engine diagnostic tool.
8	Engine Stop	21	Fault	The engine has triggered the Engine Stop Lamp condition.	Control system will not allow engine restart until power cycle. Engine may stop itself, depending on the engine option.	* Check the engine fault codes with an engine diagnostic tool. Note: The control system shuts off the engine immediately upon receiving this fault from the engine.
9	Engine	21	Fault	Engine is reporting an engine-specific fault	This is a generic engine warning fault. Engine may enforce power reduction. Other effects vary depending on the fault.	Refer to the engine fault codes in the menu system.
10	Ground Controls Engine Start Switch	27	Active at startup	Engine start button held at startup.	Engine start is disabled	Release engine start button, check DISCON overlay and ribbon cable if issue persists.
11	Platform Controls Engine Start Switch	27	Active at startup	Platform controls engine start switch active at startup.	Engine start switch will not operate.	*Verify with a voltmeter that the Engine Start Toggle Switch (TS2) is operating properly. * C33STR (BK, PCON J16-3) should be 0 V with the switch OFF and 12 V with the switch ON. *Cycle power to allow the machine to re-test the fault.

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
12	Machine CAN Bus	13	Not detected	Machine CANBUS CAN+ or CAN- shorted to supply	Machine will not be functional.	*Verify with a voltmeter that CAN+ and CAN- lines are not shorted. *Measure the resistance between D82CAN+ (YL, TCON J5-3) and D81CAN- (GR, TCON J5-4); should be between 55 and 65 ohms. *D81CAN+ (YL, TCON J5-3) should be between 1V and 5V. *D81CAN- (GR, TCON J5-4) should be between 1V and 5V. *Remove short(s) and cycle power.
13	Jib Up/Down Switch	27	Active at startup	Toggle switch input was detected as HIGH at start-up.	The Jib Up or Down functions will not operate.	*Verify with a voltmeter that the Jib Up/Down Toggle Switch (TS8) is operating properly. * C43JU (BL, PCON J17-6) should be 0 V with the switch OFF and 12 V with the switch held in the UP position. * C44JD (BL/BK, PCON J17-8) should be 0 V with the switch OFF and 12 V with the switch held in the DOWN position. *Cycle power to allow the machine to re-test the fault.
15	Foot Switch	27	Active at startup	Foot switch input was detected as HIGH at start-up, or the footswitch was held for too long and caused the footswitch timer to elapse.	Machine will not be functional.	* Release foot switch and cycle power. * If problem persists, with a voltmeter verify that C24FS (WH, PCON J12-2) is LOW when foot switch is released and 12 V when held down.
		28	Timeout	Foot switch depressed too long without action	Machine will not be functional.	
17	Aux Pump Coil	11	Shorted to supply voltage	TCON detected that C27AUX circuit is shorted to 12V.	Machine will not be functional.	*Check C27AUX (RD, TCON J3-3) wiring. *With a voltmeter verify that C27AUX (RD, TCON J3-3) is 0 V when the Auxiliary Power toggle switch is OFF and 12 V when switch is ON. *Verify the terminals of the APU solenoid have a resistance greater than 5 ohms. If not, replace the solenoid.
19	Aux Enable Switch	27	Active at startup	Aux button held at startup.	Machine will not be functional.	Release Aux button, check DISCON overlay and ribbon cable if issue persists.

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
20	Pri Boom Ext / Ret Joystick	11	Shorted to supply voltage	Circuit C163PER is above 4.5 V.	Boom Extend and Retract functions will not operate.	* Leave joystick in neutral position and cycle power. * If fault persists, with a voltmeter check that C163PER (BL/WH, PCON J18-4) is at 2.5V +/- 0.2 V at startup and does not go above 4.5 V or below 0.5 V when fully stroked in each direction. * Replace joystick if voltage check fails.
		27	Active at startup	Circuit C163PER voltage was not within the startup voltage range of 2.5 V +/- 0.2 V.		
21	Pri Boom Up / Down Joystick	11	Shorted to supply voltage	Circuit C164PUD is above 4.5 V.	Boom Up and Down functions will not operate.	* Leave joystick in neutral position and cycle power. * If fault persists, with a voltmeter check that C164PUD (RD/WH, PCON J18-3) is at 2.5V +/- 0.2 V at startup and does not go above 4.5 V or below 0.5 V when fully stroked in each direction. * Replace joystick if voltage check fails.
		27 28	Active at startup	Circuit C164PUD voltage was not within the startup voltage range of 2.5 V +/- 0.2 V.		
22	Pri Boom Up coil	11	Shorted to supply voltage	TCON detected that C1PBU circuit is shorted to 12 V.	Boom Up function will not operate. Boom Down will also not operate if short is to 12 V.	* With a voltmeter, measure Y22. Proper resistance should be between 5 - 60 ohms. Replace Y22 if resistance is outside of those values. * Verify wiring for C1PBU (RD, TCON J4-10) and check for open or short circuits.
23	Pri Boom Down Coil	11	Shorted to supply voltage	TCON detected that C2PBD circuit is shorted to 12 V or GND.	Boom Up and Down functions will not operate.	* With a voltmeter, measure Y21. Proper resistance should be between 5 - 60 ohms. Replace Y21 if resistance is outside of those values. * Verify wiring for C2PBD (RD/BK, TCON J4-9) and check for open or short circuits. * Verify resistance between Recovery Module pins J37-2 and J37-11 is zero ohms.

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
24	Pri Boom Angle	13	Not detected	No CAN messages have been received by the TCON from the Angle Sensor.	Machine limited to recovery functions	* Verify sensor power P85PWR (GR, TCON J6-7) is at 12 V while the system is running. * Verify the sensor is receiving power. Unplug the sensor and verify 12 V across pins 2 (+) and 1 (-) on the harness connector. * Verify the CAN bus wiring. With the sensor still unplugged, measure resistance across harness connector pins 3 and 4. If resistance is not between 55 and 65 ohms, check CAN wire harnesses. * Replace sensor if necessary.
		14	Cross check failed	The two internal sensing channels inside the Angle Sensor have registered a significant difference.		* Calibrate boom angle sensor. * Check boom angle sensor mount, ensure all fasteners are tightened to spec. * Replace sensor if issue persists.
		17	Not calibrated	Sensor has been replaced and requires calibration, or calibration in memory has been erased.		Run the boom angle calibration process and verify fault is no longer active.
		19	Value out of expected range	Min or max boom angle out of expected range. Crosscheck limit switch disagrees with boom angle sensor.	Machine limited to recovery functions.	* Verify that with the boom stowed C68LS (BL, TCON J7-3) is 12V and that the boom is making contact with the LST3S limit switch arm. * Verify that with the boom above 12 degrees C68LS (BL, TCON J7-3) is 0V and that the boom is not making contact with the LST3S limit switch arm. * Check LST3S limit switch for physical damage. * Check bearings in rotary base housing. * Calibrate boom angle sensor.
25	Cable tension	21	Fault	Boom cable tension limit switch LSB2S indicates that the tension is loose on the boom cables.	Boom Up and Extend functions will not operate.	* Return boom to stowed position as quickly as possible. * Verify both boom cables are properly tensioned and that LST3S is properly centered in the bracket. * Verify LSB2S (C59CNK, BL/WH, TCON J7-5) operates properly. Signal should normally be 12 V. If cable tension is loose, signal should be 0 V.

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
26	Pri Boom Ext Coil	11	Shorted to supply voltage	TCON detected that C7PBE circuit is shorted to 12 V or GND.	Boom Extend and Retract functions will not operate.	* With a voltmeter, measure Y26. Proper resistance should be between 5 - 60 ohms. Replace Y26 if resistance is outside of those values. * Verify wiring for C7PBE (BK, TCON J4-7) and check for open or short circuits.
27	Pri Boom Ret Coil	11	Shorted to supply voltage	TCON detected that C8PBR circuit is shorted to 12 V or GND.	Boom Extend and Retract functions will not operate.	* With a voltmeter, measure Y25. Proper resistance should be between 5 - 60 ohms. Replace Y25 if resistance is outside of those values. * Verify wiring for C8PBR (BK/WH, TCON J4-8) and check for open or short circuits. * Verify resistance between Recovery Module pins J37-5 and J37-8 is zero ohms.
35	Ground Control Button	27	Active at startup	Buttons were held at system start, or are shorted on the ground control panel.	The function associated with the stuck button will not operate.	*Release the ground control buttons and cycle power. *If problem persists, open ground control box and open the Display Controller access door, being careful to keep water out. Verify the ribbon cables are properly seated and dry. *Re-seat and/or replace ribbon cables if necessary. *Replace ground control overlay.
36	Jib Up Coil	11	Shorted to supply voltage	PCON detected that V43JU circuit is shorted to 12 V or GND.	Jib Up function will not operate.	* With a voltmeter, measure Y68. Proper resistance should be between 5 - 60 ohms. Replace Y68 if resistance is outside of those values. * Verify wiring for V43JU (BL, PCON J12-8) and check for open or short circuits.
37	Jib Down Coil	11	Shorted to supply voltage	PCON detected that V44JD circuit is shorted to 12 V or GND.	Jib Down function will not operate.	* With a voltmeter, measure Y69. Proper resistance should be between 5 - 60 ohms. Replace Y69 if resistance is outside of those values. * Verify wiring for V44JD (BL/BK, PCON J12-9) and check for open or short circuits.

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
40	Jib Select Coil	11	Shorted to supply voltage	PCON detected V18PRR is short to supply	Jib and platform functions will not operate	*With a voltmeter, measure Y18. Proper resistance should be between 5 - 60 ohms. Replace Y18 if resistance is outside of those values. *Verify wiring for V18PRR (GR/BK, PCON J12-4) and check for short circuits.
41	Turntable Rotate Joystick	11	Shorted to supply voltage	Circuit C165TTR is above 4.5 V.	Turntable Rotate functions will not operate.	* Leave joystick in neutral position and cycle power. * If fault persists, with a voltmeter check that C165TTR (WH/RD, PCON J18-2) is at 2.5V +/- 0.2 V at startup and does not go above 4.5 V or below 0.5 V when fully stroked in each direction. * Replace joystick if voltage check fails.
		27	Active at startup	Circuit C165TTR voltage was not within the startup voltage range of 2.5 V +/- 0.2 V.		
42	Turntable Rotate CW Coil	11	Shorted to supply voltage	TCON detected that C4TRL circuit is shorted to 12 V or GND.	Turntable Rotate functions will not operate.	* With a voltmeter, measure Y24. Proper resistance should be between 5 - 60 ohms. Replace Y24 if resistance is outside of those values. * Verify wiring for C4TRL (WH, TCON J3-12) and check for open or short circuits.
43	Turntable Rotate CCW Coil	11	Shorted to supply voltage	TCON detected that C5TRR circuit is shorted to 12 V or GND.	Turntable Rotate functions will not operate.	* With a voltmeter, measure Y23. Proper resistance should be between 5 - 60 ohms. Replace Y23 if resistance is outside of those values. * Verify wiring for C5TRR (WH/BK, TCON J3-1) and check for open or short circuits.
44	Drive Enable Toggle Switch	27	Active at startup	Toggle switch input was detected as HIGH at start-up.	Drive Enable switch will not operate, possibly preventing drive while the turntable is rotated.	*Verify with a voltmeter that the Drive Enable Toggle Switch (TS15) is operating properly. * C143DEL (BL/RD, PCON J16-7) should be 0 V with the switch OFF and 12 V with the switch held in the ON position. *Cycle power to allow the machine to re-test the fault.

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
45	Platform Level Toggle Switch	27	Active at startup	Toggle switch input was detected as HIGH at start-up.	Platform Level Up or Down functions may not operate, depending on which direction the switch fault occurred.	*Verify with a voltmeter that the Platform Level Up/Down Toggle Switch (TS9) is operating properly. * C14PLU (OR, PCON J17-2) should be 0 V with the switch OFF and 12 V with the switch held in the UP position. * C15PLD (OR/BK, PCON J17-3) should be 0 V with the switch OFF and 12 V with the switch held in the DOWN position. *Cycle power to allow the machine to re-test the fault.
46	Platform Rotate CW Coil	11	Shorted to supply voltage	PCON detected that V17PRL circuit is shorted to 12 V or GND.	Platform Rotate CW function will not operate.	* With a voltmeter, measure Y70. Proper resistance should be between 5 - 60 ohms. Replace Y70 if resistance is outside of those values. * Verify wiring for V17PRL (GR, PCON J12-5) and check for open or short circuits.
47	Platform Rotate CCW Coil	11	Shorted to supply voltage	PCON detected that V18PRR circuit is shorted to 12 V or GND.	Platform Rotate CCW function will not operate.	* With a voltmeter, measure Y71. Proper resistance should be between 5 - 60 ohms. Replace Y71 if resistance is outside of those values. * Verify wiring for V18PRR (GR/BK, PCON J12-4) and check for open or short circuits.
51	Drive Joystick	11	Shorted to supply voltage	Circuit C160DRV is above 4.5 V.	Drive Forward and Reverse functions will not operate.	* Leave joystick in neutral position and cycle power. * If fault persists, with a voltmeter check that C160DRV (WH/RD, PCON J18-10) is at 2.5 V +/- 0.2 V at startup and does not go above 4.5 V or below 0.5 V when fully stroked in each direction. * Replace joystick if voltage check fails.
		27	Active at startup	Circuit C160DRV voltage was not within the startup voltage range of 2.5 V +/- 0.2 V.		

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
52	Drive Forward Pump Coil	11	Shorted to supply voltage	TCON detected that C30FWD circuit is shorted to 12 V or GND.	Drive Forward, Brake Release and possibly Engine will not operate.	* With a voltmeter, measure Y6. Proper resistance should be between 5 - 60 ohms. Replace Y6 if resistance is outside of those values. * Verify wiring for C30FWD (WH, TCON J2-6) and check for open or short circuits.
53	Drive Reverse Pump Coil	11	Shorted to supply voltage	TCON detected that C31REV circuit is shorted to 12 V or GND.	Drive Reverse, Brake Release and possibly Engine will not operate.	* With a voltmeter, measure Y5. Proper resistance should be between 5 - 60 ohms. Replace Y5 if resistance is outside of those values. * Verify wiring for C31REV (WH/BK, TCON J2-7) and check for open or short circuits.
54	Brake Release Coil	11	Shorted to supply voltage	TCON detected that C32BRK circuit is shorted to 12 V or GND.	Machine will not be functional.	* With a voltmeter, measure Y2. Proper resistance should be between 5 - 60 ohms. Replace Y2 if resistance is outside of those values. * Verify wiring for C32BRK (WH/RD, TCON J2-9) and check for open or short circuits.
55	High Drive Speed Coil	11	Shorted to supply voltage	TCON detected that C29MS circuit is shorted to 12 V or GND.	Drive functions will not operate.	* With a voltmeter, measure Y27. Proper resistance should be between 5 - 60 ohms. Replace Y27 if resistance is outside of those values. * Verify wiring for C29MS (RD/WH, TCON J2-8) and check for open or short circuits.
56	Telematics Device	13	Not detected	M7 telematics device not detected.	Information only	Cycle power. If fault persists, check telematics ready connector wiring. If there are no wiring issues replace M7.

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
61	Steer Joystick	11	Shorted to supply voltage	Circuit C159STD or C159STA is above 4.5 V.	Drive and steer functions will not operate.	* Leave joystick in neutral position and cycle power. * If fault persists, identify which circuit is the steer circuit. Check C159STA (BL/WH, PCON J18-11) for a joystick steer option. Check C159STD (BL/RD, PCON J18-9) for the rocker joystick steer option. With a voltmeter check that the circuit is at 2.5 V +/- 0.2 V at startup and does not go above 4.5 V or below 0.5 V when fully stroked in each direction. * Replace joystick if voltage check fails.
		27	Active at startup	Circuit C159STD or C159STA voltage was not within the startup voltage range of 2.5 V +/- 0.2 V.		
62	Prox Override Toggle Switch	27	Active at startup	Toggle switch input was detected as HIGH at start-up.	The Jib Up or Down functions will not operate.	* Verify with a voltmeter that the Jib Up/Down Toggle Switch (TS8) is operating properly. * C43JU (BL, PCON J17-6) should be 0 V with the switch OFF and 12 V with the switch held in the UP position. * C44JD (BL/BK, PCON J17-8) should be 0 V with the switch OFF and 12 V with the switch held in the DOWN position. Cycle power to allow the machine to re-test the fault.
68	Ground Function Enable Button	27	Active at startup	Button was held at system start, or is shorted on the ground control panel.	The function associated with the stuck button will not operate.	* Release the ground control buttons and cycle power. * If problem persists, open ground control box and open the Display Controller access door, being careful to keep water out. Verify the ribbon cables are properly seated and dry. * Re-seat and/or replace ribbon cables if necessary. * Replace ground control overlay.

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
69	AC Generator Output	11	Shorted to supply voltage	TCON detected that C45GEN circuit is shorted to 12 V.	Generator function will not operate.	* With a voltmeter, measure Y29. Proper resistance should be between 5 - 60 ohms. Replace Y29 if resistance is outside of those values. * Verify wiring for C45GEN (GR/WH, TCON J3-2) and check for open or short circuits.
72	Lift Guard Contact Alarm Switch	14	Cross check failed	PCON detected that inputs from SW30 (contact alarm proximity sensor) are the same voltage.	Boom Up and Extend functions will not operate.	* Using a voltmeter, verify that the voltage on C248CA (OR/BK, PCON J12-1) is always the opposite of C247CA (OR, PCON J12-11). * Check wiring for open or short circuits. * Verify SW30 is receiving power and ground by verifying 12 V across C31 pins 1 and 3. * Replace SW30 if necessary.
		21	Fault			
73	Engine Oil Pressure	12	Sensor value too high	Engine oil pressure sender is shorted to 12V.	Engine will not operate.	* With a voltmeter, check that C25PSR (WH/BK, TCON J6P9) is not shorted to 12V and voltage is between 0.1-11.5V * Replace sender if necessary
		24	Too low	The engine oil pressure is measured as too low, potentially damaging the engine.	Engine may not attempt to start until system power cycle. Engine may be turned off by control system immediately.	* Stop engine as soon as possible. * Verify the engine oil level. * Verify the engine oil pump. * Power on the machine and start the engine. At the ground controls display, view the engine oil pressure gauge for proper operation. * If engine uses an analog oil pressure sensor, verify it is sending an analog voltage on C25PSR (WH/BK, TCON J6-9).

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
74	Engine Temperature	15	Sensor value too low	Engine temperature sensor is shorted to ground	Engine will not operate	* With a voltmeter, check that C26TSR (WH/RD, TCON J6P8) is not shorted to ground and voltage is between 0.1-11.5V * Replace sensor if necessary
		23	Too high	The engine coolant temperature is measured as too high, potentially damaging the engine.	Engine will not restart until machine power is cycled. Engine may stop itself, depending on the engine option.	* Stop engine as soon as possible. * Verify the engine coolant level. * Verify the engine coolant/water pump. * Power on the machine and start the engine. At the ground controls display, view the engine temperature gauge for proper operation. * If engine uses an analog temperature sensor, verify it is sending an analog voltage on C26TSR (WH/RD, TCON J6-10).
76	Platform Load Sensor	13	Not detected	Load sensor not detected on the communications bus.	Boom Up, Extend and Turntable Rotate functions will not operate.	* Verify sensor power P26PWR (BK, PCON J13-9) is at 12 V while the system is running. * Verify 12 V across pins on C27-2 (+) and C27-3 (-) on the harness connector. * Measure resistance across C27 pins 4 and 5 on the harness connector. If resistance is not between 55 and 65 ohms, check CAN wire harnesses. * Replace sensor if necessary.
		14	Cross check failed	The two internal sensing channels inside the Load Sensor have registered a significant difference.	Boom Up, Extend and Turntable Rotate functions will not operate.	* Check that the load sensor is mounted correctly and all fasteners are tightened to spec. * Calibrate load sensor. * Replace sensor if issue persists.
		15	Value to low	Platform load sensor out of calibration	Machine limited to recovery functions	* Make sure the platform is not touching the ground or an overhead obstruction. Cycle power and check that fault is no longer reported. * Perform the platform overload calibration procedure. * Replace sensor if necessary.
		17	Not calibrated	Platform load sensor out of calibration		* Calibrate load sensor. * Replace sensor if issue persists.
		21	Fault	Platform load sensor out of calibration		* Check that the load sensor is mounted correctly and all fasteners are tightened to spec. * Calibrate load sensor. * Replace sensor if issue persists.



Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
78	Platform Controller (PCON)	13	Not detected	TCON has not detected the PCON primary processor on the communications bus.	Controls from the platform will not operate.	* Verify PCON is receiving power. Check for 12 V on J15-2 (+) and J15-1 (-). * Verify PCON is receiving ESTOP signals. With ground ESTOP pulled out and keyswitch set to Platform, verify 12 V is reaching PCON J15-10. * Verify CAN bus is functional. Pull out connector J15 from the PCON and with a voltmeter set to resistance verify that pins 3 and 4 on the harness connector measure between 110 and 130 Ohms. Check wiring and boom cable for open or short circuits if resistance check fails. * Replace PCON if necessary.
		29	Software version mismatch	PCON software is out of date.	Machine limited to recovery functions	Download latest software from firmware site and update machine.
79	Display Controller (DISCON)	13	Not detected	TCON has not detected the DISCON on the communications bus.	Controls from the ground will not operate.	* Verify DISCON is operating by checking that the display on the front is operating properly. * If there is no display, try to power up the DISCON by setting the keyswitch to Ground, pull out the ground ESTOP and press the Ground Function Enable button. The display should start. * If there is still no display, check DISCON power on J10 pins 2 (+) and 1 (-). * If DISCON operates but fault persists, check the CAN bus wiring. Disconnect J10 and with a voltmeter set to resistance verify that the resistance between pins 3 and 4 on the harness connector are between 55 and 65 ohms. Check CAN wiring if resistance check fails. * Replace DISCON if required.
		29	Software version mismatch	DISCON software is out of date.	Machine limited to recovery functions	Download latest software from firmware site and update machine.

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
80	Turntable Controller (TCON)	13	Not detected	The DISCON has not detected the TCON on the CAN bus.	Machine limited to recovery functions	* Verify TCON is receiving power. Check for 12 V on J5-2 (+) and J5-1 (-). * Verify TCON is receiving ESTOP signals. With ground ESTOP pulled out and keyswitch set to Ground, verify 12 V is reaching TCON J5-9 and J5-12. * Verify CAN bus is functional. Pull out connector J5 from the TCON and with a voltmeter set to resistance verify that pins 3 and 4 on the harness connector measure between 55 and 65 Ohms. Check wiring and boom cable for open or short circuits if resistance check fails. * Replace TCON if necessary.
		29	Software version mismatch	TCON software is out of date		Download latest software from firmware site and update machine.
82	Overload	21	Recovery	This is a flag that indicates that the platform was overloaded and that auxiliary function enable was used while overloaded.	Advisory condition. Machine is fully functional.	Power up the machine in service mode and clear the flag in the service menu on the ground control display.
83	Disabled	21	By owner	The telematics module has locked out machine operation.	Machine will not be functional, or have limited functionality, depending on the telematics option.	Call machine owner to re-enable machine functions.
84	Chassis Angle	13	Not detected	The tilt sensor S80 has not been detected on the CAN bus.	Boom Up, Extend and Drive functions will not operate.	* Verify S80 is receiving power. Disconnect sensor connector and verify harness connector pins 1 and 2 measure 12 V. * Verify CAN bus wiring. With a voltmeter set to resistance, check the resistance between pins 3 and 4 on the harness connector (not the sensor's integrated connector). Resistance should be between 55 and 65 ohms. Check CAN wiring if resistance check fails. * Replace sensor if required.
		17	Not calibrated	Tilt sensor not calibrated	Machine limited to recovery functions	Calibrate tilt sensor.

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
85	Chassis Angle Safety	13	Not detected	Tilt sensor not calibrated	Machine limited to recovery functions	Calibrate tilt sensor.
88	Function Enable Coil	11	Shorted to supply voltage	TCON detected that C155PCE circuit is shorted to 12 V or GND.	Only Auxiliary Power is available.	* With a voltmeter, measure Y74. Proper resistance should be between 5 - 60 ohms. Replace Y74 if resistance is outside of those values. * Verify wiring for C155PCE (OR/RD, TCON J2-5) and check for open or short circuits.
89	Fuel Level Sensor	12	Value to high	Fuel sensor shorted to B+	Information only	With a voltmeter, check voltage on C65FLI (BL/WH, TCON J7-11). The voltage should be between 1V and 5V. Check for shorts to B+.
		15	Value to low	Fuel sensor shorted to GND		With a voltmeter, check voltage on C65FLI (BL/WH, TCON J7-11). The voltage should be between 1V and 5V. Check for shorts to ground.
92	Turntable Rotate Flow Coil	11	Shorted to supply voltage	TCON detected that C6TRF circuit is shorted to 12 V or GND.	Turntable Rotate functions will not operate.	* With a voltmeter, measure Y13. Proper resistance should be between 5 - 60 ohms. Replace Y13 if resistance is outside of those values. * Verify wiring for C6TRF (WH/RD, TCON J4-1) and check for open or short circuits.

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
94	Primary Boom Length	13	Not detected	No CAN messages have been received by the TCON from the Length Sensor.	Boom Up, Extend and Turntable Rotate functions will not operate.	* Verify sensor power P85PWR (GR, TCON J6-7) is at 12 V while the system is running. * Verify the sensor is receiving power. Unplug the sensor and verify 12 V across pins 1 (+) and 2 (-) on the harness connector. * Verify the CAN bus wiring. With the sensor still unplugged, measure resistance across harness connector pins 3 and 4. If resistance is not between 55 and 65 ohms, check CAN wire harnesses. * Replace sensor if necessary.
		14	Cross check failed	The two internal sensing channels inside the Length Sensor have registered a significant difference.	Machine limited to recovery functions	* Ensure that the boom angle sensor cord is hooked up to the bracket on the boom and that the cord extends and retracts smoothly. * Calibrate boom length sensor. * Replace sensor if necessary.
		17	Not calibrated	Boom length sensor not calibrated.		Calibrate boom length sensor.
		19	Value out of expected range	Min or max length out of expected range. Crosscheck limit switch disagrees with length sensor.		* Verify that with the boom retracted C24LS (OR/RD TCON J7-4) is 12V and that the boom is making contact with the LSB1RS limit switch arm. * Verify that when the boom is extended at least 10" C24LS (OR/RD, TCON J7-4) is OV and that the boom is not making contact with the LSB1RS limit switch arm. * Check LSB1RS limit switch for physical damage. * Calibrate length sensor.
99	Primary Boom Extend / Retract Flow Coil	11	Shorted to supply voltage	TCON detected that C9PEF circuit is shorted to 12 V or GND.	Boom Extend and Retract functions will not operate.	* With a voltmeter, measure Y15. Proper resistance should be between 5 - 60 ohms. Replace Y15 if resistance is outside of those values. * Verify wiring for C9PEF (BK/RD, TCON J4-5) and check for open or short circuits. * Verify resistance between Recovery Module pins J37-6 and J37-7 is zero ohms.

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
100	Platform Level Up Coil	11	Shorted to supply voltage	PCON detected that V14PLU circuit is shorted to 12 V or GND.	Platform Level Up and possibly Level Down functions will not operate.	* With a voltmeter, measure Y20. Proper resistance should be between 5 - 60 ohms. Replace Y20 if resistance is outside of those values. * Verify wiring for V14PLU (OR, PCON J12-6) and check for open or short circuits.
101	Platform Level Down Coil	11	Shorted to supply voltage	PCON detected that V15PLD circuit is shorted to 12 V or GND.	Platform Level Down and possibly Level Up functions will not operate.	* With a voltmeter, measure Y19. Proper resistance should be between 5 - 60 ohms. Replace Y19 if resistance is outside of those values. * Verify wiring for V15PLD (OR/BK, PCON J12-7) and check for open or short circuits.
110	Platform Rotate Toggle Switch @PC	27	Active at startup	Toggle switch input was detected as HIGH at start-up.	Platform Rotate Left or Right will not operate, depending on which side of the switch was faulted.	*Verify with a voltmeter that the Platform Rotate Toggle Switch (TS7) is operating properly. * C17PRL (GR, PCON J16-4) and C18PRR (GR/BK, PCON J17-5) should be 12 V when the switch is pressed in their respective directions. *Cycle power to allow the machine to re-test the fault.
116	Model Configuration	31	Setup invalid.	Machine is not fully configured or set to wrong model.	Machine will not operate.	Navigate to Settings -> Model Configuration and ensure model, engine, tires, and region are set and correct.
141	Boom	10	Safety Envelope Exceeded	The system has determined that the platform position has exceeded the safety envelope boundary.	All functions will not operate except for the recovery procedure.	Use Auxiliary Power and the boom retract and/or down functions to bring the platform back within the operational envelope.
150	Sensor Power P26PWR	11	Shorted to supply voltage	PCON detected that P26PWR circuit is shorted to 12 V or GND.	Boom Up and Extend functions will not operate.	* Turn on machine and verify sensor power P26PWR (BK, PCON J13-9) is at 12 V. * Inspect wiring for open or shorts circuits.
152	Primary Boom Up/Down Flow Coil	11	Shorted to supply voltage	TCON detected that C3PBF circuit is shorted to 12 V or GND.	Boom Up and Down functions will not operate.	* With a voltmeter, measure Y12. Proper resistance should be between 5 - 60 ohms. Replace Y12 if resistance is outside of those values. * Verify wiring for C3PBF (RD/WH, TCON J4-4) and check for open or short circuits. * Verify resistance between Recovery Module pins J37-3 and J37-10 is zero ohms.

Control System Fault Codes

ID	Component	ID	Name	Cause	Effect	Solution
160	LF Steer (L or R) Coil	11	Shorted to supply voltage	TCON detected that C37STL circuit is shorted to 12 V or GND.	Steer Left function will not operate.	* With a voltmeter, measure Y3. Proper resistance should be between 5 - 60 ohms. Replace Y3 if resistance is outside of those values. * Verify wiring for C37STL (BL/BK, TCON J4-11) and check for open or short circuits.
161	RF Steer (L or R) Coil	11	Shorted to supply voltage	TCON detected that C36STR circuit is shorted to 12 V or GND.	Steer Right function will not operate.	* With a voltmeter, measure Y4. Proper resistance should be between 5 - 60 ohms. Replace Y4 if resistance is outside of those values. * Verify wiring for C36STR (BL, TCON J4-12) and check for open or short circuits.
185	ESTOP Source	11	Shorted to supply voltage	TCON detected that ESTOP 1 is at 12 V but ESTOP 3 is at 0 V.	ESTOP circuit cannot complete, preventing machine functions.	* Using a voltmeter and with the system turned OFF, check the resistance between TCON J5 pins 9 and 10. Replace TCON if resistance is more than 5 ohms. * Check P79PWR (BK, TCON J5-9) and P23PWR (RD, TCON J5-10) for open or short circuits.
186	ESTOP Return	11	Shorted to supply voltage	Mismatched signals detected on ESTOP Return circuits. Either: * PCON ESTOP 2 and ESTOP 4 do not match, or * TCON ESTOP 2 and ESTOP 4 do not match.	ESTOP circuit cannot complete, preventing machine functions.	* Inspect PCON J15 pins 11 and 12 for open or short circuits. Using a voltmeter, set to diode test mode and make sure the diode tests ok with the anode at pin 12 and cathode at pin 11. If test fails, replace the PCON. * Inspect TCON J5 pins 11 and 12 for open or short circuits. Using a voltmeter, set to diode test mode and make sure the diode tests ok with the anode at pin 12 and cathode at pin 11. If test fails, replace the PCON. * Check P23RET (BK) for shorts to ground or power.
187	TCON Bank power	13	Not detected	Power plug not connected or loose. Internal TCON component failure.	TCON outputs will not turn on.	* Check J1, Bank Power, on TCON is making a good connection. * Verify pin 1 is grounded * Verify pins 2, 3, 4 are 12V * Replace TCON
188	PCON Bank power	13	Not detected	Power plug not connected or loose. Internal TCON component failure.	PCON outputs will not turn on.	* Check J11, Bank Power, on PCON is making a good connection. * Verify pin 1 is grounded * Verify pins 3, 4 are 12V * Replace PCON

Engine Fault Codes

Engine Fault Codes

Some engine options have their own Electronic Control Module that contains more detailed fault information that is specific to the engine. This system can be accessed by using the engine's dedicated service tool. This service tool may be connected to the Engine Service Connector, located to the right of the battery tray.

The ground controls display and fault menu will show basic engine faults, such as low oil pressure and high coolant temperature. It will also show faults for engine warning and engine stop, which correspond to the engine warning lamp and engine stop lamp indicators that the engine may require.

More advanced faults specific to the engine will require the use of a corresponding engine service tool.

Deutz TD 2.9 L4 Engine Fault Codes

SPN = Suspect Parameter Number

FMI = Failure Mode Identifier

KWP = Keyword Protocol

SPN	FMI	KWP	Description
51	3	1019	EGR-Valve, short circuit to battery
51	3	1024	Position sensor error of actuator EGR-Valve; signal range check high
51	3	1226	EGR-Valve; short circuit to battery (A02)
51	3	1227	EGR-Valve; short circuit to battery (A67)
51	4	1020	EGR-Valve; short circuit to ground
51	4	1025	Position sensor error actuator EGR-Valve; signal range check low
51	4	1228	EGR-Valve; short circuit to ground (A02)
51	4	1229	EGR-Valve; short circuit to ground (A67)
1	4	1232	Actuator error EGR-Valve; Voltage below threshold
51	5	1015	Actuator error EGR-Valve; signal range check low
51	5	1017	Actuator EGR-Valve; open load
51	5	1023	Actuator error EGR-Valve; signal range check low
51	5	1223	Actuator EGR-Valve; open load
51	6	1014	Actuator error EGR-Valve; signal range check high
51	6	1022	Actuator error EGR-Valve; signal range check high
51	6	1224	Actuator EGR-Valve; over current
51	6	1230	Actuator error EGR-Valve; Overload by short-circuit
51	7	1016	Actuator position for EGR-Valve not plausible

SPN	FMI	KWP	Description
51	11	1231	Actuator error EGR-Valve; Power stage over temp due to high current
51	12	1018	Actuator EGR-Valve; powerstage over temperature
51	12	1021	Mechanical actuator defect EGR-Valve
51	12	1225	Actuator EGR-Valve; over temperature
94	1	474	Low fuel pressure; warning threshold exceeded
94	1	475	Low fuel pressure; shut off threshold exceeded
94	3	472	Sensor error low fuel pressure; signal range check high
94	4	473	Sensor error low fuel pressure; signal range check low
97	3	464	Sensor error water in fuel; signal range check high
97	4	465	Sensor error water in fuel; signal range check low
97	12	1157	Water in fuel level prefilter; maximum value exceeded
100	0	734	High oil pressure; warning threshold exceeded
100	0	735	High oil pressure; shut off threshold exceeded
100	1	736	Low oil pressure; warning threshold exceeded
100	1	737	Low oil pressure; shut off threshold exceeded
100	3	732	Sensor error oil pressure; signal range check high
100	4	733	Sensor error oil pressure sensor; signal range check low
102	2	88	Charged air pressure above warning threshold
102	2	89	Charged air pressure above shut off threshold
102	4	777	Sensor error charged air press.; signal range check low

Deutz TD 2.9 L4 Engine Fault Codes

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SPN	FMI	KWP	Description
105	0	996	High charged air cooler temperature; warning threshold exceeded
105	0	997	High charged air cooler temperature; shut off threshold exceeded
105	3	994	Sensor error charged air temperature; signal range check high
105	4	995	Sensor error charged air temperature; signal range check low
108	3	412	Sensor error ambient air press.; signal range check high
108	4	413	Sensor error ambient air press.; signal range check low
110	0	98	High coolant temperature; warning threshold exceeded
110	0	99	High coolant temperature; shut off threshold exceeded
110	3	96	Sensor error coolant temp.; signal range check high
110	4	97	Sensor error coolant temp.; signal range check low
111	1	101	Coolant level too low
132	11	1	Air flow sensor load correction factor exceeding the maximum drift limit; plausibility error
132	11	2	Air flow sensor load correction factor exceeding drift limit; plausibility error
132	11	3	Air flow sensor low idle correction factor exceeding the maximum drift limit
132	11	4	Air flow sensor load correction factor exceeding the maximum drift limit
157	3	877	Sensor error rail pressure; signal range check high
157	4	878	Sensor error rail pressure; signal range check low

SPN	FMI	KWP	Description
168	0	1180	Physical range check high for battery voltage
168	1	1181	Physical range check low for battery voltage
168	2	47	High battery voltage; warning threshold exceeded
168	2	48	Low battery voltage; warning threshold exceeded
168	3	45	Sensor error battery voltage; signal range check high
168	4	46	Sensor error battery voltage; signal range check low
171	3	417	Sensor error environment temperature; signal range check high
171	4	418	Sensor error environment temperature; signal range check low
172	0	1182	Physical range check high for intake air temperature
172	1	1183	Physical range check low for intake air temperature
172	2	9	Sensor ambient air temperature; plausibility error
172	2	983	Intake air sensor; plausibility error
172	3	981	Sensor error intake air; signal range check high
172	4	982	Sensor error intake air sensor; signal range check low
174	0	481	High low fuel temperature; warning threshold exceeded
174	0	482	High Low fuel temperature; shut off threshold exceeded
175	0	740	Physical range check high for oil temperature
175	0	745	High oil temperature; warning threshold exceeded
175	0	746	High oil temperature; shut off threshold exceeded
175	1	741	Physical range check low for oil temperature

Deutz TD 2.9 L4 Engine Fault Codes

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SPN	FMI	KWP	Description
175	2	738	Sensor oil temperature; plausibility error
175	2	739	Sensor oil temperature; plausibility error oil temperature too high
175	3	743	Sensor error oil temperature; signal range check high
175	4	744	Sensor error oil temperature; signal range check low
190	0	389	Engine speed above warning threshold (FOC-Level 1)
190	2	421	Offset angle between crank- and camshaft sensor is too large
190	8	419	Sensor camshaft speed; disturbed signal
190	8	422	Sensor crankshaft speed; disturbed signal
190	11	390	Engine speed above warning threshold (FOC-Level 2)
190	12	420	Sensor camshaft speed; no signal
190	12	423	Sensor crankshaft speed; no signal
190	14	391	Engine speed above warning threshold (Overrun Mode)
190	14	1222	Camshaft- and Crankshaft speed sensor signal not available on CAN
411	0	791	Physical range check high for differential pressure Venturiunit (EGR)
411	1	792	Physical range check low for differential pressure Venturiunit (EGR)
411	3	795	Sensor error differential pressure Venturiunit (EGR); signal range check high
411	4	381	Physical range check low for EGR differential pressure
411	4	796	Sensor error differential pressure Venturiunit (EGR); signal range check low

SPN	FMI	KWP	Description
412	3	1007	Sensor error EGR cooler downstream temperature; signal range check high
412	4	1008	Sensor error EGR cooler downstream temperature; signal range check low
520	9	306	Timeout Error of CAN-Receive-Frame TSC1TR; Setpoint
597	2	49	Break lever mainswitch and break lever redundancy switch status not plausible
624	3	971	SVS lamp; short circuit to batt.
624	4	972	SVS lamp; short circuit to grd.
624	5	969	SVS lamp; open load
624	12	970	SVS lamp; powerstage over temperature
630	12	376	Access error EEPROM memory (delete)
630	12	377	Access error EEPROM memory (read)
630	12	378	Access error EEPROM memory (write)
639	14	84	CAN-Bus 0 "BusOff-Status"
651	3	580	Injector 1 (in firing order); short circuit
651	4	586	High side to low side short circuit in the injector 1 (in firing order)
651	5	568	Injector 1 (in firing order); interruption of electric connection
652	3	581	Injector 2 (in firing order); short circuit
652	4	587	High side to low side short circuit in the injector 2 (in firing order)
652	5	569	Injector 2 (in firing order); interruption of electric connection
653	3	582	Injector 3 (in firing order); short circuit
653	4	588	High side to low side short circuit in the injector 3 (in firing order)
653	5	570	Injector 3 (in firing order); interruption of electric connection

Deutz TD 2.9 L4 Engine Fault Codes

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SPN	FMI	KWP	Description
654	3	583	Injector 4 (in firing order); short circuit
654	4	589	High side to low side short circuit in the injector 4 (in firing order)
654	5	571	Injector 4 (in firing order); interruption of electric connection
676	11	543	Cold start aid relay error.
676	11	544	Cold start aid relay open load
677	3	956	Starter relay high side; short circuit to battery
677	3	960	Starter relay low side; short circuit to battery
677	4	957	Starter relay high side; short circuit to ground
677	4	961	Starter relay low side; short circuit to ground
677	5	958	Starter relay; no load error
677	12	959	Starter relay; powerstage over temperature
703	3	426	Engine running lamp; short circuit to battery
703	4	427	Engine running lamp; short circuit to ground
703	5	424	Engine running lamp; open load
703	12	425	Engine running lamp; powerstage over temperature
729	5	545	Cold start aid relay open load
729	12	547	Cold start aid relay; over temperature error
898	9	305	Timeout Error of CAN-Receive-Frame TSC1TE; Setpoint
1079	13	946	Sensor supply voltage monitor 1 error (ECU)
1080	13	947	Sensor supply voltage monitor 2 error (ECU)

SPN	FMI	KWP	Description
1109	2	121	Engine shut off demand ignored
1136	0	1398	Physikal range check high for ECU temperature
1136	1	1399	Physikal range check low for ECU temperature
1136	3	1400	Sensor error ECU temperature; signal range check high
1136	4	1401	Sensor error ECU temperature; signal range check low
1176	3	849	Sensor error pressure sensor upstream turbine; signal range check high
1176	4	850	Sensor error pressure sensor downstream turbine; signal range check high
1180	0	1193	Physical range check high for exhaust gas temperature upstream turbine
1180	0	1460	Turbocharger Wastegate CAN feedback; warning threshold exceeded
1180	0	1462	Exhaust gas temperature upstream turbine; warning threshold exceeded
1180	1	1194	Physical range check low for exhaust gas temperature upstream turbine
1180	1	1461	Turbocharger Wastegate CAN feedback; shut off threshold exceeded
1180	1	1463	Exhaust gas temperature upstream turbine; shut off threshold exceeded
1180	3	1067	Sensor error exhaust gas temperature upstream turbine; signal range check high
1180	11	1066	Sensor exhaust gas temperature upstream turbine; plausibility error
1188	2	1414	Wastegate; status message from ECU missing

Deutz TD 2.9 L4 Engine Fault Codes

SPN = Suspect Parameter Number

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SPN	FMI	KWP	Description
1188	7	1415	Wastegate actuator; blocked
1188	11	1411	Wastegate actuator; internal error
1188	11	1412	Wastegate actuator; EOL calibration not performed correctly
1188	11	1416	Wastegate actuator; over temperature (> 145°C)
1188	11	1417	Wastegate actuator; over temperature (> 135°C)
1188	11	1418	Wastegate actuator; operating voltage error
1188	13	1413	Wastegate actuator calibration deviation too large, recalibration required
1231	14	85	CAN-Bus 1 "BusOff-Status"
1235	14	86	CAN-Bus 2 "BusOff-Status"
1237	2	747	Override switch; plausibility error
1322	12	610	Too many recognized misfires in more than one cylinder
1323	12	604	Too many recognized misfires in cylinder 1 (in firing order)
1324	12	605	Too many recognized misfires in cylinder 2 (in firing order)
1325	12	606	Too many recognized misfires in cylinder 3 (in firing order)
1326	12	607	Too many recognized misfires in cylinder 4 (in firing order)
2659	0	1524	Physical range check high for EGR exhaust gas mass flow
2659	1	1525	Physical range check low for EGR exhaust gas mass flow
2659	2	1523	Exhaust gas recirculation AGS sensor; plausibility error

SPN	FMI	KWP	Description
2659	2	1527	AGS sensor temperature exhaust gas mass flow; plausibility error
2659	12	1526	Exhaust gas recirculation; AGS sensor has "burn off" not performed
2797	4	1337	Injector diagnostics; timeout error of short circuit to ground measurement cyl. Bank 0
2798	4	1338	Injector diagnostics; timeout error of short circuit to ground measurement cyl. Bank 1
2798	4	1339	Injector diagnostics; short circuit to ground monitoring Test in Cyl. Bank 0
2798	4	1340	Injector diagnostics; short circuit to ground monitoring Test in Cyl. Bank 1
3224	2	127	DLC Error of CAN-Receive-Frame AT11G1 NOX Sensor (SCR-system upstream cat; DPF-system downstream cat); length of frame incorrect
3224	9	128	Timeout Error of CAN-Receive-Frame AT11G1; NOX sensor upstream
3248	4	1047	Sensor error particle filter downstream temperature; signal range check low
3699	2	1616	DPF differential pressure sensor and a further sensor or actuator CRT system defective
3699	2	1617	Temperature sensor us. and ds. DOC simultaneously defect
3699	14	1615	Maximum stand-still-duration reached; oil exchange required
4765	0	1039	Physical range check high for exhaust gas temperature upstream (DOC)
4765	1	1042	Physical range check low for exhaust gas temperature upstream (DOC)

Deutz TD 2.9 L4 Engine Fault Codes

SPN = Suspect Parameter Number

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SPN	FMI	KWP	Description
4766	0	1029	Physical range check high for exhaust gas temperature downstream (DOC)
4766	1	1032	Physical range check low for exhaust gas temperature downstream (DOC)
4768	2	1036	Sensor exhaust gas temperature upstream (DOC); plausibility error
4768	3	1044	Sensor error exhaust gas temperature upstream (DOC); signal range check high
4768	4	1045	Sensor error exhaust gas temperature upstream (DOC) signal range check low
4769	2	1026	Sensor exhaust gas temperature downstream (DOC); plausibility error
4769	3	1034	Sensor error exhaust gas temperature downstream (DOC); signal range check high
4769	4	1035	Sensor error exhaust gas temperature downstream (DOC); signal range check low
523006	3	34	Controller mode switch; short circuit to battery
523006	4	35	Controller mode switch; short circuit to ground
523008	1	648	Manipulation control was triggered
523008	2	649	Timeout error in Manipulation control
523009	9	825	Pressure Relief Valve (PRV) reached maximum allowed opening count
523009	10	833	Pressure relief valve (PRV) reached maximum allowed open time

SPN	FMI	KWP	Description
523212	9	171	Timeout Error of CAN-Receive-Frame ComEngPrt; Engine Protection
523216	9	198	Timeout Error of CAN-Receive-Frame PrHtEnCmd; pre-heat command, engine command
523240	9	179	Timeout CAN-message FunModCtl; Function Mode Control
523350	4	565	Injector cylinder-bank 1; short circuit
523352	4	566	Injector cylinder-bank 2; short circuit
523354	12	567	Injector powerstage output defect
523470	2	826	Pressure Relief Valve (PRV) forced to open; performed by pressure increase
523470	2	827	Pressure Relief Valve (PRV) forced to open; performed by pressure shock
523470	7	876	Maximum rail pressure in limp home mode exceeded (PRV)
523470	11	831	The PRV can not be opened at this operating point with a pressure shock
523470	11	832	Rail pressure out of tolerance range
523470	12	828	Open Pressure Relief Valve (PRV); shut off condition
523470	12	829	Open Pressure Relief Valve (PRV); warning condition
523470	14	830	Pressure Relief Valve (PRV) is open
523550	12	980	T50 start switch active for too long
523601	13	948	Sensor supply voltage monitor 3 error (ECU)
523603	9	126	Timeout Error of CAN-Receive-Frame AMB; Ambient Temperature Sensor

Deutz TD 2.9 L4 Engine Fault Codes

SPN = Suspect Parameter Number

FMI = Failure Mode Identifier

KWP = Keyword Protocol

SPN	FMI	KWP	Description
523605	9	300	Timeout Error of CAN-Receive-Frame TSC1AE; Traction Control
523606	9	301	Timeout Error of CAN-Receive-Frame TSC1AR; Retarder
523612	12	387	Internal software error ECU; injection cut off
523612	12	612	Internal ECU monitoring detection reported error
523612	12	613	Internal ECU monitoring detection reported error
523612	12	614	Internal ECU monitoring detection reported error
523612	12	615	Internal ECU monitoring detection reported error
523612	12	616	Internal ECU monitoring detection reported error
523612	12	617	Internal ECU monitoring detection reported error
523612	12	618	Internal ECU monitoring detection reported error
523612	12	619	Internal ECU monitoring detection reported error
523612	12	620	Internal ECU monitoring detection reported error
523612	12	621	Internal ECU monitoring detection reported error
523612	12	623	Internal ECU monitoring detection reported error
523612	12	624	Internal ECU monitoring detection reported error
523612	12	625	Internal ECU monitoring detection reported error
523612	12	627	Internal ECU monitoring detection reported error
523612	12	628	Internal ECU monitoring detection reported error

SPN	FMI	KWP	Description
523612	12	637	Internal ECU monitoring detection reported error
523612	12	1170	Internal software error ECU
523612	14	973	Softwarereset CPU SWReset_0
523612	14	974	Softwarereset CPU SWReset_1
523612	14	975	Softwarereset CPU SWReset_2
523613	0	856	Maximum positive deviation of rail pressure exceeded (RailMeUn0)
523613	0	857	Maximum positive deviation of rail pressure in metering unit exceeded (RailMeUn1)
523613	0	858	Railsystem leakage detected (RailMeUn10)
523613	0	859	Maximum negative deviation of rail pressure in metering unit exceeded (RailMeUn2)
523613	0	860	Negative deviation of rail pressure second stage (RailMeUn22)
523613	0	862	Maximum rail pressure exceeded (RailMeUn4)
523613	1	861	Minimum rail pressure exceeded (RailMeUn3)
523613	2	864	Setpoint of metering unit in overrun mode not plausible
523615	3	594	Metering unit (Fuel-System); short circuit to battery highside
523615	3	596	Metering unit (Fuel-System); short circuit to battery low side
523615	4	595	Metering unit (Fuel-System); short circuit to ground high side
523615	4	597	Metering Unit (Fuel-System); short circuit to ground low side
523615	5	592	Metering unit (Fuel-System); open load
523615	12	593	Metering unit (Fuel-System); powerstage over temperature
523619	2	488	Physical range check high for exhaust gas temperature upstream (SCR-CAT)
523698	11	122	Shut off request from supervisory monitoring function

Deutz TD 2.9 L4 Engine Fault Codes

SPN = Suspect Parameter Number

FMI = Failure Mode Identifier

KWP = Keyword Protocol

SPN	FMI	KWP	Description
523717	12	125	Timeout Error of CAN-Transmit-Frame AmbCon; Weather environments
523718	3	1488	SCR mainrelay; short circuit to battery (only CV56B)
523718	4	1489	SCR mainrelay; short circuit to ground (only CV56B)
523718	5	1486	SCR mainrelay; open load (only CV56B)
523718	12	1487	SCR mainrelay; powerstage over temperature (only CV56B)
523766	9	281	Timeout Error of CAN-Receive-Frame Active TSC1AE
523767	9	282	Timeout Error of CAN-Receive-Frame Passive TSC1AE
523768	9	283	Timeout Error of CAN-Receive-Frame Active TSC1AR
523769	9	284	Timeout Error of CAN-Receive-Frame Passive TSC1AR
523770	9	285	Timeout Error of CAN-Receive-Frame Passive TSC1DE
523776	9	291	Timeout Error of CAN-Receive-Frame TSC1TE - active
523777	9	292	Passive Timeout Error of CAN-Receive-Frame TSC1TE; Setpoint
523778	9	293	Active Timeout Error of CAN-Receive-Frame TSC1TR
523779	9	294	Passive Timeout Error of CAN-Receive-Frame TSC1TR

SPN	FMI	SPN	Description
523788	12	299	Timeout Error of CAN-Transmit-Frame TrbCH; Status Wastegate
523793	9	202	Timeout Error of CAN-Receive-Frame UAA10; AGS sensor service message
523794	9	203	Timeout Error of CAN-Receive-Frame UAA11; AGS sensor data
523895	13	559	Check of missing injector adjustment value programming (IMA) injector 1 (in firing order)
523896	13	560	check of missing injector adjustment value programming (IMA) injector 2 (in firing order)
523897	13	561	check of missing injector adjustment value programming (IMA) injector 3 (in firing order)
523898	13	562	check of missing injector adjustment value programming (IMA) injector 4 (in firing order)
523910	6	1261	Air Pump; over current
523913	3	74	Sensor error glow plug control diagnostic line voltage; signal range check high
523913	4	75	Sensor error glow plug control diagnostic line voltage; signal range check low
523914	3	78	Glow plug control; short circuit to battery
523914	4	79	Glow plug control; short circuit to ground
523914	5	76	Glow plug control; open load
523914	5	1216	Glow plug control release line; short circuit error
523914	11	1217	Glow plug control; internal error
523914	12	77	Glow plug control; powerstage over temperature
523919	2	1378	Sensor air pump airpressure; plausibility error
523920	2	1379	Sensor exhaust gas back pressure burner; plausibility error

Deutz TD 2.9 L4 Engine Fault Codes

SPN = Suspect Parameter Number

FMI = Failure Mode Identifier

KWP = Keyword Protocol

SPN	FMI	KWP	Description
523922	7	1262	Burner Shut Off Valve; blocked closed
523922	7	1264	Burner Shut Off Valve; blocked closed
523929	0	109	Fuel Balance Control integrator injector 1 (in firing order); maximum value exceeded
523929	1	115	Fuel Balance Control integrator injector 1 (in firing order); minimum value exceeded
523930	0	110	Fuel Balance Control integrator injector 2 (in firing order); maximum value exceeded
523930	1	116	Fuel Balance Control integrator injector 2 (in firing order); minimum value exceeded
523931	0	111	Fuel Balance Control integrator injector 3 (in firing order); maximum value exceeded
523931	1	117	Fuel Balance Control integrator injector 3 (in firing order); minimum value exceeded
523932	0	112	Fuel Balance Control integrator injector 4 (in firing order); maximum value exceeded
523932	1	118	Fuel Balance Control integrator injector 4 (in firing order); minimum value exceeded
523935	12	168	Timeout Error of CAN-Transmit-Frame EEC3VOL1; Engine send messages
523936	12	169	Timeout Error of CAN-Transmit-Frame EEC3VOL2; Engine send messages
523946	0	1158	Zero fuel calibration injector 1 (in firing order); maximum value exceeded

SPN	FMI	KWP	Description
523946	1	1164	Zero fuel calibration injector 1 (in firing order); minimum value exceeded
523947	0	1159	Zero fuel calibration injector 2 (in firing order); maximum value exceeded
523947	1	1165	Zero fuel calibration injector 2 (in firing order); minimum value exceeded
523948	0	1160	Zero fuel calibration injector 3 (in firing order); maximum value exceeded
523948	1	1166	Zero fuel calibration injector 3 (in firing order); minimum value exceeded
523949	0	1161	Zero fuel calibration injector 4 (in firing order); maximum value exceeded
523949	1	1167	Zero fuel calibration injector 4 (in firing order); minimum value exceeded
523960	0	1011	Physical range check high for EGR cooler downstream temp.
523960	0	1458	High exhaust gas temperature EGR cooler downstream; warning threshold exceeded
523960	1	1012	Physical range check low for EGR cooler downstream temp.
523960	1	1459	High exhaust gas temperature EGR cooler downstream; shut off threshold exceeded
523980	14	1187	Bad quality of reduction agent detected
523981	11	918	Urea-tank without heating function (heating phase)
523982	0	360	Powerstage diagnosis disabled; high battery voltage
523982	1	361	Powerstage diagnosis disabled; low battery voltage



Deutz TD 2.9 L4 Engine Fault Codes

SPN = Suspect Parameter Number

FMI = Failure Mode Identifier

KWP = Keyword Protocol

SPN	FMI	KWP	Description
523988	3	1245	Charging lamp; short circuit to battery
523988	4	1246	Charging lamp; short circuit to ground
523988	5	1243	Charging lamp; open load
523988	12	1244	Charging lamp; over temp.
523998	4	1327	Injector cylinder bank 2 slave; short circuit
523999	12	1328	Injector powerstage output Slave defect
524014	1	1254	Air pressure glow plug flush line; below limit
524016	2	1259	Amount of air is not plausible to pump speed
524016	2	1260	Calculated amount of air is not plausible to HFM reading
524016	11	1258	HFM sensor; electrical fault
524021	11	1263	Burner fuel line pipe leak behind Shut Off Valve
524024	11	1302	Deviation of the exhaust gas temp. setpoint to actual value downstream (DOC) too high
524028	2	1431	CAN message PROEGRActr; plausibility error
524029	2	1432	Timeout Error of CAN-Receive-Frame ComEGRActr - exhaust gas recirculation positioner
524030	7	1440	EGR actuator; internal error
524031	13	1441	EGR actuator; calibration error

SPN	FMI	KWP	Description
524032	2	1442	EGR actuator; status message EGRCust is missing
524033	7	1443	EGR actuator; due to overload in Save Mode
524034	3	1438	Disc separator; short circuit to battery
524034	4	1439	Disc separator; short circuit to ground
524034	5	1436	Disc Separator; open load
524034	12	1437	Disc Separator; powerstage over temperature
524035	12	1341	Injector diagnostics; time out error in the SPI communication
524057	2	1505	Electric fuel pump; fuel pressure build up error
524097	9	1663	Timeout error of CAN-Transmit-Frame DPFBnAirPmpCtl
524098	9	1664	Timeout error of CAN-Transmit-Frame ComDPFBnPT
524099	9	1665	Timeout error of CAN-Transmit-Frame ComDPFC1
524100	9	1666	Timeout error of CAN-Transmit-Frame ComDPFHisDat
524101	9	1667	Timeout error of CAN-Transmit-Frame ComDPFTstMon

Deutz TD 2.9 L4 Engine Fault Codes

SPN = Suspect Parameter Number

FMI = Failure Mode Identifier

KWP = Keyword Protocol

SPN	FMI	KWP	Description
524102	9	1674	Timeout error of CAN-Receive-Frame ComRxDPFBrnAirPmpCtl
524103	9	1675	Timeout error of CAN-Receive-Frame ComRxDPFBrnAirPmp
524104	9	1676	Timeout error of CAN-Receive-Frame ComRxDPFCtl
524105	9	1668	Timeout error of CAN-Transmit-Frame ComEGRMsFlw
524106	9	1677	Timeout error of CAN-Receive-Frame ComRxEGRMsFlw1
524107	9	1678	Timeout error of CAN-Receive-Frame ComRxEGRMsFlw2
524108	9	1669	Timeout error of CAN-Transmit-Frame ComEGRTVActr
524109	9	1679	Timeout error of CAN-Receive-Frame ComRxEGRTVActr
524110	9	1670	Timeout error of CAN-Transmit-Frame ComETVActr
524111	9	1680	Timeout error of CAN-Receive-Frame ComRxETVActr
524112	9	1671	Timeout ComITVActr
524113	9	1681	Timeout error of CAN-Receive-Frame ComRxITVActr

SPN	FMI	KWP	Description
524114	9	1659	Timeout error of CAN-Transmit-Frame A1DOC
524115	9	1660	Timeout error of CAN-Transmit-Frame AT1S
524116	9	1661	Timeout error of CAN-Transmit-Frame SCR2
524117	9	1662	Timeout error of CAN-Transmit-Frame SCR3
524118	9	1672	Timeout error of CAN-Receive-Frame ComRxCM1
524119	9	1673	Timeout error of CAN-Receive-Frame ComRxCustSCR3
524120	9	1682	Timeout error of CAN-Receive-Frame ComRxSCRHtDiag
524121	9	1683	Timeout error of CAN-Receive-Frame ComRxTrbChActr
524122	9	1684	Timeout error of CAN-Receive-Frame ComRxUQSens
524123	9	1685	Timeout error of CAN-Receive-Frame ComSCRHtCtl
524124	9	1686	Timeout error of CAN-Receive-Frame ComTxAT1IMG
524125	9	1687	Timeout error of CAN-Receive-Frame ComTxTrbChActr

Deutz TCD 2.2 L3 Engine Fault Codes

DTC = Diagnostic Trouble Code

FMI = Failure Mode Identifier

SPN = Suspect Parameter Number

DTC	SPN	FMI	Description
1000	98	2	Engine oil level sensor internal error. Sensor reports error. Open/short transducer.
1001	98	31	Engine oil level out of range. Level low, high, foaming.
1002	98	31	Oil sensor voltage out of range. <8,5V ±0,5V; >16,5V ±0,5V.
1003	98	2	Oil sensor invalid sensor status.
1004	98	31	Oil sensor temperature out of range.
1005	98	14	Oil sensor is broken or disconnected.
1021	100	3	Oil pressure voltage above normal or shorted to high.
1022	100	4	Oil pressure voltage below normal or shorted to low.
1025	100	1	Low oil pressure. Warning threshold exceeded.
1026	100	1	Low oil pressure. Shut off threshold exceeded.
1043	107	0	Air filter differential pressure. Air filter clogged.
1071	411	2	Engine exhaust gas recirculation. Pressure does not change between engine operating points.
1072	411	0	Engine exhaust gas recirculation. Pressure above normal operational range.
1073	411	1	Engine exhaust gas recirculation. Pressure below normal operational range.
1074	411	2	Engine exhaust gas recirculation. Negative measured differential pressure.
1075	411	2	Engine exhaust gas recirculation. Positive measured differential pressure.

DTC	SPN	FMI	Description
1077	411	3	Engine exhaust gas recirculation. Signal value above maximum limit.
1078	411	4	Engine exhaust gas recirculation. Signal value below maximum limit.
1079	108	0	Ambient air pressure sensor above normal operational range.
1080	108	1	Ambient air pressure sensor below normal operational range.
1081	108	15	Fault check max signal range violated for ambient air pressure sensor.
1082	108	17	Fault check min signal range violated for ambient air pressure sensor.
1083	108	2	Ambient air pressure sensor error by component self diagnosis.
1084	3720	0	DPF ash load above normal operational range.
1086	3734	0	DPF soot load exceeded. Remove filter level.
1087	4781	14	DPF soot load exceeded shut off level.
1088	4781	0	DPF soot load exceeded warning level.
1089	4781	16	DPF. Too much standstill time in short time interval.
1090	10156	0	DPF. The standstill-regeneration mode time exceeds the short-limit.
1091	3735	16	DPF. Standstill required and no successful standstill longer than escalation threshold. Moderately severe.
1092	3735	0	DPF. Standstill required and no successful standstill longer than escalation threshold. Most severe.
1093	4766	1	DOC. Regeneration temperature in standstill main phase not reached.
1102	171	2	Ambient air temperature shows a deviation from expected value at cold start conditions.

Deutz TCD 2.2 L3 Engine Fault Codes

DTC = Diagnostic Trouble Code

FMI = Failure Mode Identifier

SPN = Suspect Parameter Number

DTC	SPN	FMI	Description
111	102	0	Engine intake manifold pressure above normal operational range.
1114	102	1	Engine intake manifold pressure below normal operational range.
1115	102	3	Intake manifold pressure sensor voltage above normal or shorted to high.
1116	102	4	Intake manifold pressure sensor voltage below normal or shorted to low.
1118	102	1	Intake manifold pressure below normal operational range.
1121	102	2	DFC for signal variation check for pressure sensor of the intake manifold.
1122	102	0	Intake air pressure valve sensor, warning condition exceeded.
1123	102	1	Intake air pressure valve sensor, shutoff condition exceeded.
1124	1209	2	Engine exhaust pressure turbine upstream differs from ambient pressure while engine not running.
1125	1209	15	Engine exhaust pressure turbine upstream above upper limit.
1126	1176	1	Engine turbocharger compressor intake pressure below normal operational range.
1127	1209	2	Engine exhaust pressure turbine upstream tuck check failed. Pressure does not change between engine operating points.
1130	1209	3	Engine exhaust pressure sensor voltage above normal or shorted to high.
1131	1209	4	Engine exhaust pressure sensor voltage below normal or shorted to low.

DTC	SPN	FMI	Description
1134	3251	3	DPF voltage above normal or shorted to high.
1135	3251	4	DPF voltage below normal or shorted to low.
1136	3251	14	DPF reporting communication error.
1137	3251	14	DPF reporting data error.
1138	3251	14	DPF reporting fast channel 1 error.
1139	3251	14	DPF reporting fast channel 2 signal range error.
1149	3251	2	DPF difference pressure value not plausible.
1150	3251	0	DPF difference pressure above shut off threshold.
1151	3251	16	DPF difference pressure above warning threshold.
1152	3251	1	DPF difference pressure below shut off threshold.
1153	3251	18	DPF difference pressure below warning threshold.
1161	5571	16	Rail fuel pressure relief valve above normal operational range.
1162	5571	2	Rail fuel pressure relief valve is forced to open, perform pressure increase.
1163	5571	2	Rail fuel pressure relief valve is forced to open. Performed by pressure increase.
1164	5571	16	Rail fuel pressure relief valve is forced to open. Shutoff conditions.
1165	5571	15	Rail fuel pressure relief valve is forced to open. Warning conditions.
1166	5571	0	Open rail fuel pressure relief valve was detected.
1167	5571	2	Unexpected opening of the rail fuel pressure relief valve.
1168	5571	2	Successful rail fuel pressure relief valve opening cannot be ensured.
1169	5571	13	Averaged rail fuel pressure after valve opening is outside the expected tolerance range.

Deutz TCD 2.2 L3 Engine Fault Codes

DTC = Diagnostic Trouble Code

FMI = Failure Mode Identifier

SPN = Suspect Parameter Number

DTC	SPN	FMI	Description
1170	5571	16	Open time of rail fuel pressure relief valve for wear out monitoring had exceeded.
1171	94	1	Fuel pressure build up during engine start not successful.
1172	1347	5	Electrical fuel pump current below normal or open circuit.
1174	1347	3	Electrical fuel pump voltage above normal or shorted to high.
1175	1347	4	Electrical fuel pump voltage below normal or shorted to low.
119	1231	14	CAN Bus 2 off Error for Application CAN.
1190	7103	13	Rail fuel pressure below set point, speed-dependent threshold exceeded.
1191	7103	13	Rail fuel pressure metering unit. Fuel quantity balance is disrupted.
1194	7103	13	Negative rail fuel pressure governor deviation at zero delivery by metering unit.
1195	7103	1	Rail fuel pressure value is below minimum rail pressure threshold.
1197	7103	0	Maximum rail fuel pressure exceeded.
1198	7103	2	Set point of fuel metering unit in overrun mode not plausible.
120	639	14	CAN Bus 1 off Error for Power train CAN.
1200	5357	14	Shut-off due to undershoot of minimum rail pressure.
1202	157	0	Maximum rail pressure exceeded in limp home mode.
1208	157	3	Engine fuel injector metering rail pressure voltage above normal or shorted to high.

DTC	SPN	FMI	Description
1209	157	4	Engine fuel injector metering rail pressure voltage below normal or shorted to low.
121	520252	2	Wrong checksum in the CAN message EAT Control.
1212	629	12	ECU. Keep alive error during runtime at an external device.
1213	629	12	ECU. Keep alive error during initialization phase at an external device.
1215	629	12	ECU. Read diagnosis error for non volatile memory.
1216	629	12	ECU. Write diagnosis error for non volatile memory.
1218	629	12	ECU. Stack memory threshold overrun.
1219	629	12	ECU. Observation counter irregular switch off counter triggered by engine running.
122	4207	2	TSC1 message checksum fault.
123	4207	2	TSC1 message checksum fault.
1233	5826	15	Emission control system operator inducement level 1 severity above normal operational range.
1235	5826	0	Emission control system operator inducement level 2 severity above normal operational range.
1236	5826	14	Emission control system operator pre-trigger inducement level 2 severity.
124	4207	2	TSC1 message checksum fault.
125	4207	2	TSC1 message checksum fault.
1274	91	3	Accelerator pedal sensor position 1 voltage above normal or shorted to high.
1275	2623	3	Accelerator pedal 1, channel 2 voltage above normal or shorted to high.
1276	29	3	Accelerator pedal 2 voltage above normal or shorted to high.

Deutz TCD 2.2 L3 Engine Fault Codes

DTC = Diagnostic Trouble Code

FMI = Failure Mode Identifier

SPN = Suspect Parameter Number

DTC	SPN	FMI	Description
1277	2625	3	Accelerator pedal 2, channel 2 voltage above normal or shorted to high.
1280	91	4	Accelerator pedal sensor position 1 voltage below normal or shorted to low.
1281	2623	4	Accelerator pedal 1, channel 2 voltage below normal or shorted to low.
1282	29	4	Accelerator pedal 2 voltage below normal or shorted to low.
1283	2625	4	Accelerator pedal 2, channel 2 voltage below normal or shorted to low.
1289	3509	14	Failure of sensor supply voltage 1 from ECU.
1290	3509	0	Sensor supply voltage 1 from ECU above normal operational range.
1291	3509	6	Sensor supply voltage 1 from ECU current above normal or grounded circuit.
1292	3509	1	Sensor supply voltage 1 from ECU below normal operational range.
1293	3510	14	Failure of sensor supply voltage 2 from ECU.
1294	3510	0	Sensor supply voltage 2 from ECU above normal operational range.
1295	3510	6	Sensor supply voltage 2 from ECU current above normal or grounded circuit.
1296	3510	1	Sensor supply voltage 2 from ECU below normal operational range.
1306	677	3	Engine starter motor relay voltage above normal or shorted to high.
1307	677	4	Engine starter motor relay voltage below normal or shorted to low.

DTC	SPN	FMI	Description
1308	677	5	Engine starter motor relay current below normal or shorted to low.
1310	677	3	Engine starter motor relay voltage above normal or shorted to high.
1311	677	4	Engine starter motor relay voltage below normal or shorted to low.
1323	91	11	Accelerator pedal position 1. Possible error between APP1 and APP2 or APP1 and idle switch.
1326	29	11	Accelerator Pedal 2 Position. Possible error between APP1 and idle switch.
1346	1041	14	Start signal indicator. Terminal 50 was operated too long.
1354	105	0	Engine intake manifold 1 temperature data above normal operational range. Warning threshold exceeded.
1355	105	0	Engine intake manifold 1 temperature above normal operational range. Shutoff threshold exceeded.
1357	1136	0	Engine ECU temperature above normal operational range. Most severe.
1358	1136	1	Engine ECU temperature below normal operational range. Most severe.
1359	1136	15	Engine ECU temperature above normal operational range. Least severe.
1360	1136	17	Engine ECU temperature below normal operational range. Least severe.
1361	1136	2	Engine ECU temperature fault check.
1362	412	15	Engine exhaust gas recirculation temperature above normal operational range.

Deutz TCD 2.2 L3 Engine Fault Codes

DTC = Diagnostic Trouble Code

FMI = Failure Mode Identifier

SPN = Suspect Parameter Number

DTC	SPN	FMI	Description
1363	412	17	Engine exhaust gas recirculation temperature below normal operational range.
1364	412	3	Engine exhaust gas recirculation temperature voltage above normal or shorted to high.
1365	412	4	Engine exhaust gas recirculation temperature voltage below normal or shorted to low.
1372	51	5	Engine throttle valve 1, position 1 current below normal or open circuit.
1375	51	3	Engine throttle valve 1, position 1 voltage above normal or shorted to high. Short circuit to battery 1.
1376	51	3	Engine throttle valve 1, position 1 voltage above normal or shorted to high. Short circuit to battery 2.
1377	51	4	Engine throttle valve 1, position 1 voltage below normal or shorted to low. Short circuit to ground 1.
1378	51	4	Engine throttle valve 1, position 1 voltage below normal or shorted to low. Short circuit to ground 2.
1379	51	6	Engine throttle valve 1, position 1 current above normal or grounded circuit.
1382	51	7	Engine throttle valve 1 position 1 mechanical system not responding or out of adjustment. Valve stuck closed.
1383	51	7	Engine throttle valve 1 position 1 mechanical system not responding or out of adjustment. Valve stuck open.
1391	51	3	Engine throttle valve 1, position 1 voltage above normal or shorted to high.

DTC	SPN	FMI	Description
1392	51	4	Engine throttle valve 1, position 1 voltage below normal or shorted to low.
1397	105	0	Engine intake manifold 1 temperature above normal operational range.
1398	105	1	Engine intake manifold 1 temperature below normal operational range.
1399	4766	2	DOC temperature too high.
1400	4766	2	DOC temperature too low.
1401	4766	15	DOC outlet temperature above normal operational range.
1402	4766	3	DOC outlet temperature voltage above normal or shorted to high.
1403	4766	4	DOC outlet temperature voltage below normal or shorted to low.
1404	4766	2	DOC intake temperature error.
1405	4766	15	DOC Intake temperature above normal operational range.
1406	4766	3	DOC intake temperature voltage above normal or shorted to high.
1407	4766	4	DOC intake temperature voltage below normal or shorted to low.
1408	4766	2	DOC intake temperature does not change.
142	520256	9	Timeout of EAT control receive message. CAN message is not received.
144	523211	9	Timeout error of CAN receive frame EBC1.
154	523212	9	Timeout error of CAN receive frame engine protection.
1540	520254	8	The stand still regeneration mode time exceeds the long limit threshold.
1541	520255	2	Hoses connected to the dp DPF SENT sensor inverted. Swap hoses.
155	523741	14	Engine shutdown request via CAN.

Deutz TCD 2.2 L3 Engine Fault Codes

DTC = Diagnostic Trouble Code

FMI = Failure Mode Identifier

SPN = Suspect Parameter Number

DTC	SPN	FMI	Description
1587	97	0	Water in fuel level prefilter; maximum value exceeded.
188	523240	9	Timeout CAN message function mode control.
219	520253	2	Rolling counter fault CAN message EAT Control.
220	4206	2	Fault check for Rolling Counter of TSC1AE.
221	4206	2	Fault check for Rolling Counter of TSC1AR.
222	4206	2	Fault check for Rolling Counter of TSC1TE.
223	4206	2	Fault check for Rolling Counter of TSC1TR.
349	3349	0	Timeout error of CAN receive frame active TSC1AE.
350	3349	0	Timeout error of CAN receive frame passive TSC1AE.
351	3349	0	Timeout error of CAN receive frame active TSC1AR.
352	3349	0	Timeout error of CAN receive frame passive TSC1AR.
353	3349	0	Timeout error of CAN receive frame TSC1TE active.
354	3349	0	TSC1 receive timeout error. Short circuit to ground error.
355	3349	0	Timeout error of CAN receive frame TSC1TR.
356	3349	0	Passive timeout error of CAN receive frame TSC1TR.
361	3349	0	Timeout error of CAN receive frame TSC1AE. Traction Control.
363	3349	0	Timeout error of CAN receive frame TSC1AR. Retarder.
365	3349	0	Timeout error of CAN receive frame TSC1TE. Setpoint.

DTC	SPN	FMI	Description
367	3349	0	Timeout Error of CAN receive frame TSC1TR; control signal.
38	1485	3	ECM main relay voltage above normal or shorted to high.
39	1485	3	ECM main relay voltage above normal or shorted to high of actuator relay 2.
40	1485	3	ECM main relay voltage above normal or shorted to high of actuator relay 3.
41	1485	4	ECM main relay voltage below normal or shorted to low.
42	1485	4	ECM main relay voltage below normal or shorted to low of actuator relay 2.
43	1485	4	ECM main relay voltage below normal or shorted to low of actuator relay 3.
48	168	0	Battery voltage above normal operational range.
49	168	1	Battery voltage low normal operational range.
50	168	3	Battery voltage above normal or shorted to high.
51	168	4	Battery voltage above normal or shorted to low.
516	523982	0	Power stage diagnosis disabled. High battery voltage.
517	523982	1	Power stage diagnosis disabled. Low battery voltage.
52	168	0	High battery voltage. Warning threshold is exceeded.
567	27	5	Engine exhaust gas recirculation 1 valve position current below normal or open circuit.
570	27	3	Engine exhaust gas recirculation 1 valve position voltage above normal or shorted to battery 1.
571	27	3	Engine exhaust gas recirculation 1 valve position voltage above normal or shorted to battery 2.

Deutz TCD 2.2 L3 Engine Fault Codes

DTC = Diagnostic Trouble Code

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SPN = Suspect Parameter Number

DTC	SPN	FMI	Description
572	27	4	Engine exhaust gas recirculation 1 valve position voltage below normal or shorted to ground 1.
573	27	4	Engine exhaust gas recirculation 1 valve position voltage below normal or shorted to ground 2.
574	27	6	Engine exhaust gas recirculation 1 valve position current above normal or grounded circuit.
577	27	7	Engine exhaust gas recirculation 1 valve position. Mechanical system not responding or out of adjustment. Valve stuck closed.
578	27	7	Engine exhaust gas recirculation 1 valve position. Mechanical system not responding or out of adjustment. Valve stuck open.
582	5763	3	Engine exhaust gas recirculation 1, actuator 1 voltage above normal or shorted to high.
583	5763	4	Engine exhaust gas recirculation 1, actuator 1 voltage below normal or shorted to low.
586	3055	14	Internal software error ECU. Injection cut off.
587	190	0	Engine speed above warning threshold. Over speed detection in component engine protection.
588	190	0	Engine speed above warning threshold. FOC-Level 1.
589	190	0	Engine speed above warning threshold. FOC-Level 2.
590	190	0	Engine speed above warning threshold. Overrun Mode.
610	171	15	Environment temperature sensor, temperature above upper physical threshold.

DTC	SPN	FMI	Description
613	171	3	Ambient air temperature sensor voltage above normal or shorted to high.
614	171	4	Ambient air temperature sensor voltage below normal or shorted to low.
615	723	8	Camshaft speed sensor abnormal frequency or pulse width or period.
616	723	14	Camshaft sensor detection. Out of range, signal disrupted, no signal.
617	723	13	Offset angle between crank and camshaft sensor is too large.
618	4201	8	Crankshaft sensor detection. Out of range, signal disrupted, no signal.
619	4201	14	Crankshaft speed sensor. Speed detection, out of range, signal disrupted or no signal.
68	1669	14	CAN Bus ID-5. CAN Hardware registers are not updated within the expected time.
70	110	2	Engine Coolant Temperature. Data erratic, intermittent or incorrect.
709	97	3	Water in fuel indicator 1. Voltage above normal or shorted to high.
710	97	4	Water in fuel indicator 1. Voltage below normal or shorted to low.
721	94	15	Low fuel pressure system, max. physical range exceeded.
723	94	3	Engine fuel pressure sensor voltage above normal or shorted to high.
724	94	4	Engine fuel pressure sensor voltage below normal or shorted to low.
725	94	1	Low fuel pressure system, warning threshold exceeded.
726	94	1	Low fuel pressure, shut off threshold exceeded.
75	110	3	Engine coolant temperature voltage above normal or shorted to high.
76	110	4	Engine coolant temperature voltage below normal or shorted to low.

Deutz TCD 2.2 L3 Engine Fault Codes

DTC = Diagnostic Trouble Code

FMI = Failure Mode Identifier

SPN = Suspect Parameter Number

DTC	SPN	FMI	Description
77	110	0	High coolant temperature. Warning threshold exceeded.
78	110	0	Coolant temperature. System reaction initiated.
797	676	12	Engine cold start aid relay error.
798	676	5	Engine cold start aid relay current below normal or open circuit.
799	676	5	Engine cold start aid relay current below normal or open circuit.
80	411	2	Intake air massflow not in expected range.
803	676	3	Engine cold start aid relay voltage above normal or shorted to high.
805	676	4	Engine cold start aid relay voltage below normal or shorted to low.
807	2797	14	Engine fuel 1 injector, Group 1. Number of possible injections limited by the injection valve.
815	2797	4	Engine fuel 1 injector, Group 1 voltage below normal or shorted to low.
816	5358	5	Engine cylinder 1 fuel injection quantity current below normal or open circuit.
817	5359	5	Engine cylinder 2 fuel injection quantity current below normal or open circuit.
818	5360	5	Engine cylinder 3 fuel injection quantity current below normal or open circuit.
819	5361	5	Engine cylinder 4 fuel injection quantity current below normal or open circuit.
820	5362	5	Engine cylinder 5 fuel injection quantity current below normal or open circuit.

DTC	SPN	FMI	Description
821	5363	5	Engine cylinder 6 fuel injection quantity current below normal or open circuit.
822	2797	6	Engine fuel 1 injector, Group 1 current above normal or grounded circuit.
823	2798	6	Engine fuel 1 injector, Group 2 current above normal or grounded circuit.
824	5358	6	Engine cylinder 1 fuel injection quantity above normal or grounded circuit. Short circuit of the power stage low-side.
825	5359	6	Engine cylinder 2 fuel injection quantity above normal or grounded circuit. Short circuit of the power stage low-side.
826	5360	6	Engine cylinder 3 fuel injection quantity above normal or grounded circuit. Short circuit of the power stage low-side.
827	5361	6	Engine cylinder 4 fuel injection quantity above normal or grounded circuit. Short circuit of the power stage low-side.
828	5362	6	Engine cylinder 5 fuel injection quantity above normal or grounded circuit. Short circuit of the power stage low-side.
829	5363	6	Engine cylinder 6 fuel injection quantity above normal or grounded circuit. Short circuit of the power stage low-side.
83	111	1	Coolant level too low.
830	5358	6	Engine cylinder 1 fuel injection quantity above normal or grounded circuit. Short circuit between high-side and low-side of the power stage.

Deutz TCD 2.2 L3 Engine Fault Codes

DTC = Diagnostic Trouble Code

FMI = Failure Mode Identifier

SPN = Suspect Parameter Number

DTC	SPN	FMI	Description
831	5359	6	Engine cylinder 2 fuel injection quantity above normal or grounded circuit. Short circuit between high-side and low-side of the power stage.
832	5360	6	Engine cylinder 3 fuel injection quantity above normal or grounded circuit. Short circuit between high-side and low-side of the power stage.
833	5361	6	Engine cylinder 4 fuel injection quantity above normal or grounded circuit. Short circuit between high-side and low-side of the power stage.
834	5362	6	Engine cylinder 5 fuel injection quantity above normal or grounded circuit. Short circuit between high-side and low-side of the power stage.
835	5363	6	Engine cylinder 6 fuel injection quantity above normal or grounded circuit. Short circuit between high-side and low-side of the power stage.
836	105	3	Engine intake manifold temperature voltage above normal or shorted to high.
837	105	4	Engine intake manifold temperature voltage below normal or shorted to low.
838	2797	14	Engine fuel 1 injector, group 1 missing injector adjustment value programming injector 1.
839	2798	14	Engine fuel 1 injector, group 2 missing injector adjustment value programming injector 2.

DTC	SPN	FMI	Description
840	4257	14	Engine fuel 1 injector, group 3 missing injector adjustment value programming injector 3.
854	7103	5	Engine fuel metering rail pump current below normal or open circuit.
855	7103	3	Engine fuel metering rail pump voltage above normal or shorted to high. Short circuit to battery on the high side power stage.
856	7103	3	Engine fuel metering rail pump voltage above normal or shorted to high. Short circuit to battery on the low side power stage.
857	7103	4	Engine fuel metering rail pump voltage below normal or shorted to low. Short circuit to battery on the high side power stage.
858	7103	4	Engine fuel metering rail pump voltage below normal or shorted to low. Short circuit to battery on the low side power stage.
859	7103	6	Engine fuel metering rail pump current above normal or grounded circuit.
868	629	12	Function monitoring: fault of ECU ADC. Null load test pulse.
869	629	12	Function monitoring: fault of ECU ADC. Test voltage.
870	629	12	ECU. DFC to indicate ICO request from MoCSOP module.
871	91	14	Function monitoring: Monitoring of accelerator pedal position.
875	190	2	Function monitoring: Fault of engine speed check.
876	5357	2	Engine fuel injection error for multiple cylinders. Diagnostic fault check error between level 1 energizing time and level 2 information.

Deutz TCD 2.2 L3 Engine Fault Codes

DTC = Diagnostic Trouble Code

FMI = Failure Mode Identifier

SPN = Suspect Parameter Number

DTC	SPN	FMI	Description
877	5441	2	Engine fuel injection timing error for multiple cylinders.
878	5357	2	Engine fuel injection error for multiple cylinders. Diagnostic fault check to report the error due to non plausibility in ZFC.
879	523612	12	Internal recovery. Diagnosis fault check to report the error to demand for an ICO due to an error in the Pol2 shut-off.
88	598	10	Clutch switch. Abnormal rate of change.
880	523612	12	Internal recovery. Diagnosis fault check to report the error to demand for an ICO due to an error in the Pol3 efficiency factor.
881	523612	12	Internal recovery. Diagnosis fault check to report the error to demand for an ICO due to an error in change of EOM.
882	5357	2	Engine fuel injection error for multiple cylinders. Diagnosis fault check to report the error to demand for an ICO due to an error in total torque relevant quantity.
883	5357	2	Engine fuel injection error for multiple cylinders. Diagnostic fault check to report the error due to injection quantity correction.
884	5442	2	Engine fuel injection pressure error for multiple cylinders.
885	29	2	Accelerator pedal 2 position.
886	677	2	Engine starter motor relay. Function monitoring: Fault of ECU power train active.

DTC	SPN	FMI	Description
887	513	2	Actual engine percent torque. DFC to report the fault in energizing time comparison.
888	513	2	Actual engine percent torque. DFC to report in torque comparison error.
889	520250	2	Function monitoring: Error in the post-build selectable monitoring.
890	629	12	ECU. Status of the EMM alarm FCCU0 which is read out of the FCCU hardware module.
91	1109	2	Engine protection system approaching shutdown. Engine shut off demand ignored.
92	1109	14	Engine protection system approaching shutdown. Shut off request from supervisory monitoring function.
996	629	12	ECU. Diagnostic fault check to report ABE active state.
997	629	12	Function monitoring: Fault of ECU, WDA active by inquiry/response communication.
998	629	12	Function monitoring: Fault of ECU, Error Pin active suspicion of HW fault.
999	629	12	Function monitoring: Fault of ECU, WDA active by overvoltage detection.

The following DTC fault code range shares the same description. Replace the ECU.

DTC	891 - 945	Description
SPN	629	Internal ECU error.
FMI	12	